EXAM PACK

EEE475

RF and Microwave Engineering



Exam Pack

Session 2018–19 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination, 2022*

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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2018–19

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PAST PAPER SOLUTIONS

EEE475

RF and Microwave Engineering

Question Paper Solutions

Session 2018–19

4th Year 1st Semester Final Examination, 2022

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

EACH ANSWER



Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks). Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part-A · Answer any two of the following questions				
Q1	(a) Antenna Aperture and Directivity	6	11 min	☐
	(b) Beamwidths, Radiated Power and Directivity of a Given Pattern	6	11 min	
	(c) Field Regions Surrounding an Antenna	5	9 min	
	(d) Fields of an Elemental Magnetic Dipole	8	14 min	
Q2	(a) Passive Remote Sensing	5	9 min	☐
	(b) The Radar Range Equation	6	11 min	
	(c) Signal-to-Noise Ratio of a Radio Link	5	9 min	
	(d) Uniform Linear Array	9	16 min	
Q3	(a) Short Notes: Lorentz Reciprocity Theorem and Image Theory	6	11 min	☐
	(b) Line Input Impedance Using a Smith Chart	4	7 min	
	(c) Reflection Coefficient, SWR and Voltage Maximum of a Terminated Line	7	13 min	
	(d) Short-Circuited Stub Matching of a 50 Ohm Line	8	14 min	
Part-B · Answer any two of the following questions				
Q4	(a) Equivalent Circuit and Impedance of a High-Frequency Resistor			☐
	(b) Microstrip Antenna and Applications of Microstrip Line	6	11 min	
	(c) Short-Circuited Transmission Line	7	13 min	
	(d) Quarter-Wave Matching	6	11 min	
Q5	(a) ABCD Parameters of a High-Pass Filter	6	11 min	☐
	(b) Stability Circles	5	9 min	
	(c) Negative-Resistance and Feedback Oscillators	7	13 min	
	(d) Line Parameters from Open- and Short-Circuit Impedances	7	13 min	
Q6	(a) Short Notes: Poincaré Sphere and Yagi-Uda Antenna	6	11 min	☐
	(b) Power Density and Field Ratios of a Paraboloid	6	11 min	
	(c) Friis Transmission Formula	6	11 min	
	(d) Directivity of a Pyramidal Horn	7	13 min	

Formulas follow Pozar, Microwave Engineering (lines, Smith chart, matching, amplifier stability, oscillators, radar), Ludwig and Bretchko, RF Circuit Design (RF components, ABCD parameters) and Kraus, Antennas (antennas, arrays, apertures, horns, paraboloids, remote sensing).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering
4th Year 1st Semester Final Examination, 2022
Course Code: EEE-475, Session: 2018-19
Course Title: RF and Microwave Engineering

Full Marks: 100 **Credit: 3.0** **Exam Duration: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna. 6
   (b) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq \pi$. 6
   (c) Describe the field regions surrounding an antenna. 5
   (d) For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity. 8

2. (a) Describe how passive remote sensing is used to detect object. 5
   (b) Explain how the RADAR range equation is used to detect objects. 6
   (c) Show the signal to noise ratio of a radio link is related to bandwidth and system temperature. 5
   (d) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. 9

3. (a) Write short notes on.
   (i) Lorentz reciprocity theorem.
   (ii) Image theory. 6
   (b) Write the steps to determine line input impedance using Smith Chart. 4
   (c) A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line. 7
   (d) A 50 Ω transmission line is connected to a load impedance $35 - j47.5$ Ω. Find the position and length of a short circuited stub required to match the line. 8

Part-B

*[Answer any **two** of the following questions]*

4. (a) Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure [*sic*: should be curve] of the resistor. 6
   (b) What is microstrip antenna? Mention some of the applications of microstrip line. 6
   (c) Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length. 7
   (d) How is quarter wavelength transmission line used for impedance matching? 6

5. (a) For high pass filter determine ABCD parameters of a two port system. 6

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(b) What are stability circles? Explain how it is used. 5

(c) Explain the working principle of negative resistance oscillator and feedback oscillator. 7

(d) A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6\ \Omega$ and short circuit impedance $j103\ \Omega$. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input. 7

6. (a) Write short notes on the following topics.

(i) Poincare Sphere
(ii) Yagi-Uda antenna. 6

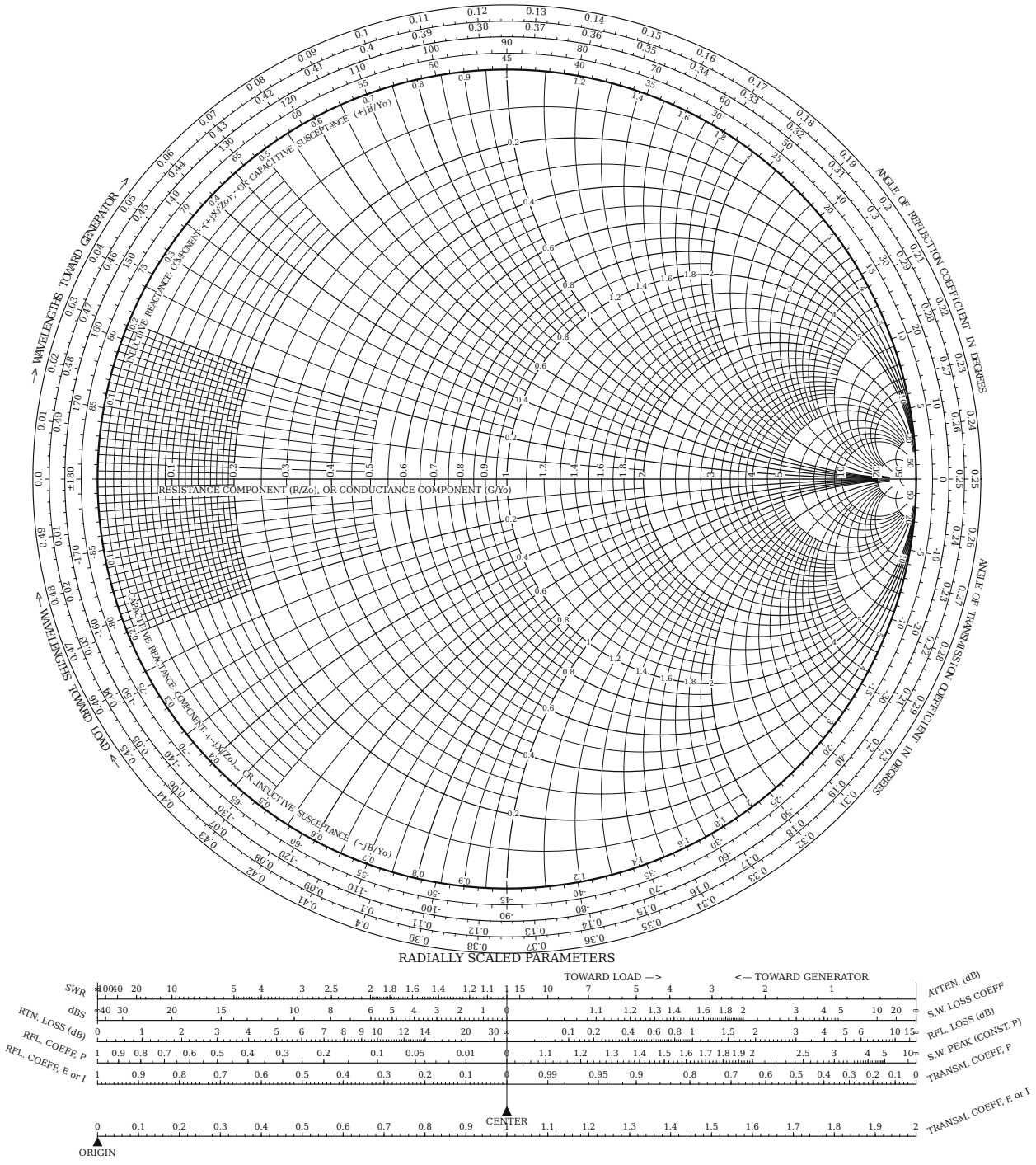
(b) For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio. 6

(c) Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them. 6

(d) For a pyramidal or rectangular horn show that directivity D is $D \approx 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right)$. 7

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Smith Chart



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1. (a) What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna. Marks: 6
   (b) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \le \theta \le \pi$, $0 \le \varphi \le \pi$. Marks: 6
   (c) Describe the field regions surrounding an antenna. Marks: 5
   (d) For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity. Marks: 8

1.(a) What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna. Marks: 6

1. Theory

1. An antenna as a "collector" (Core Idea)

A receiving antenna sits in a passing wave that carries a power density S (W/m^2). It takes some power P_r out of the wave and delivers it to its load. We can ask: *how big a window, held square to the wave, would let through the same power?* That window area is the antenna's **aperture**. For a dish or a horn it is close to the physical mouth; for a thin wire it is much bigger than the wire itself.

2. Why a bigger aperture gives a sharper beam

A wave leaving an opening of width a spreads out by an angle of about λ/a radians (diffraction, as with light through a slit). So an opening many wavelengths wide sends out a narrow beam. A narrower beam puts the power into a smaller solid angle, which is exactly what a higher directivity means.

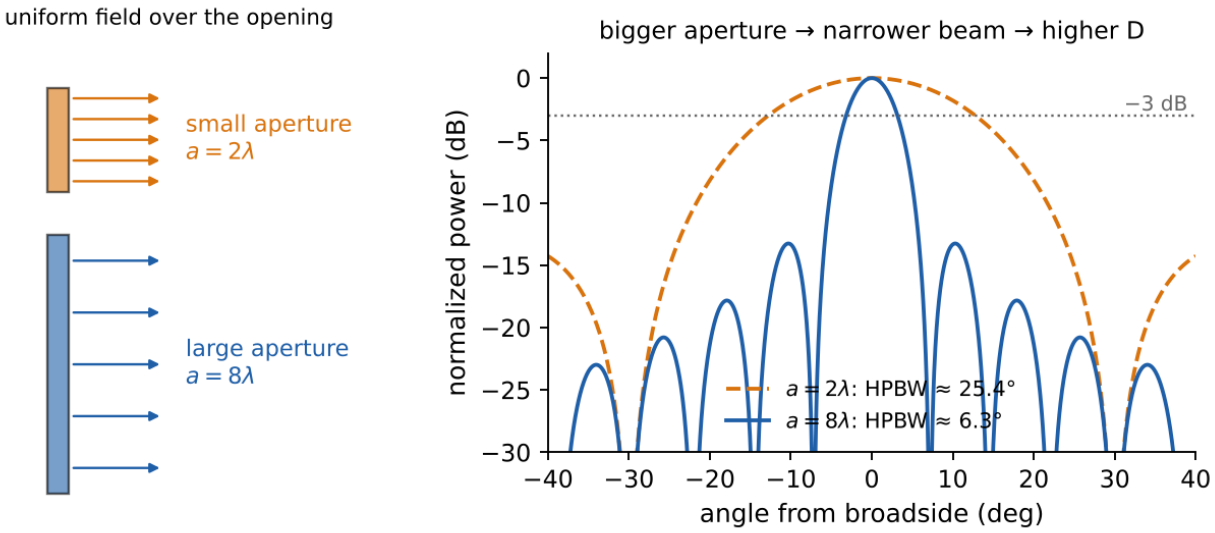


Figure 1: Bigger aperture, narrower beam

Reading the figure. Left: two openings with a uniform field: the short solid orange bar, 2λ wide, and the long solid blue bar, 8λ wide. Right: their far-field patterns in dB. The dashed orange pattern of the small aperture has a half-power beamwidth of about 25°; the solid blue one of the large aperture only about 6°.

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2. Answer

Definition: The **effective aperture** A_e of an antenna is the ratio of the power it delivers to a matched load to the power density of the incident wave:

$$A_e = \frac{P_r}{S} \quad \text{m}^2$$

The **physical aperture** A_p is the actual area of the opening (dish, horn mouth). The ratio $\varepsilon_{ap} = A_e/A_p \leq 1$ is the **aperture efficiency**.

Aperture and directivity: for any antenna

$$D = \frac{4\pi}{\lambda^2} A_e = \frac{4\pi}{\lambda^2} \varepsilon_{ap} A_p, \qquad A_e \, \Omega_A = \lambda^2$$

where:

- λ = wavelength (m), Ω_A = beam area (sr).

How the aperture is set to increase the directivity:

1. **Make the aperture large in wavelengths.** D grows with A_p/λ^2. A bigger dish or horn, or the same dish at a higher frequency, gives a narrower beam (HPBW $\approx 51^\circ \, \lambda/a$ for a uniform opening of width a; about $70^\circ \, \lambda/d$ for a dish of diameter d).
2. **Illuminate it uniformly in amplitude.** A uniform field over the whole opening gives the largest A_e ($\varepsilon_{ap} \to 1$). A field tapered towards the edges wastes part of the area (but lowers the side lobes, a trade-off).
3. **Keep the phase uniform (in phase over the opening).** Then all parts add up along the axis; a phase error (a badly shaped dish, a feed off the focus) spreads the beam and lowers D.
4. **Avoid blockage and spill-over.** A feed or struts in front of the dish, and feed power that misses the dish, reduce ε_{ap}.

Example

a 3 m dish at 10 GHz ($\lambda = 0.03$ m) with $\varepsilon_{ap} = 0.6$: $A_e = 0.6 \times 7.07 = 4.24$ m^2, so $D = 4\pi \times 4.24/0.03^2 = 5.92 \times 10^4$ (47.7 dBi). At 20 GHz the same dish gives 53.8 dBi.

3. Keyword Explanation

1. Aperture efficiency

↩ back

The fraction of the physical area that is effective, $\varepsilon_{ap} = A_e/A_p$; typically 0.5–0.8 for a dish.

4. Notes and Remarks

- A thin half-wave dipole has almost no physical area but $A_e = 0.13\lambda^2$, so "aperture" means the effective area, not the metal.

1.(b) Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \leq \theta \leq \pi$, $0 \leq \varphi \leq \pi$. Marks: 6

1. Theory

1. What is a power pattern? (Core Idea)

An antenna does not send the same power in every direction. The **radiation intensity** $U(\theta, \phi)$, in watts per steradian, tells how much power goes into each direction. A steradian is the unit of solid angle: the whole sphere is 4π sr. The **power pattern** is U as a function of direction, usually scaled so that its largest value U_m is 1.

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2. How do the angles work?

θ is measured down from the z axis, from 0 to π, and ϕ is measured round the axis in the xy plane. A small patch of solid angle in the direction (θ, ϕ) is

$$d\Omega = \sin\theta \, d\theta \, d\phi$$

The $\sin\theta$ appears because a circle of constant θ is small near the poles and large at the equator.

3. What do the four quantities mean?

- **Half-power beamwidth (HPBW):** the angle between the two directions, on either side of the peak, where the power is half of the peak.
- **Beamwidth between first nulls (FNBW):** the angle between the first zeros on either side of the peak.
- **Radiated power:** add U over every direction, $P_{rad} = \iint U \, d\Omega$.
- **Directivity:** how much stronger the peak is than the average. The average intensity is $P_{rad}/4\pi$, so

$$D = \frac{U_m}{P_{rad}/4\pi} = \frac{4\pi U_m}{P_{rad}} = \frac{4\pi}{\Omega_A}$$

where $\Omega_A = P_{rad}/U_m$ is the **beam area**: the solid angle the beam would fill if all the power went out at the peak intensity.

2. Answer

Given: $U(\theta, \phi) = \cos^2\theta \sin\phi$ for $0 \le \theta \le \pi$, $0 \le \phi \le \pi$, and zero elsewhere.

Assumption: the pattern has no constant in front, so take $U_m = 1$ W/sr. The peak is on the axis, at $\theta = 0$ or π, in the plane $\phi = 90°$. The beamwidths are read in that plane.

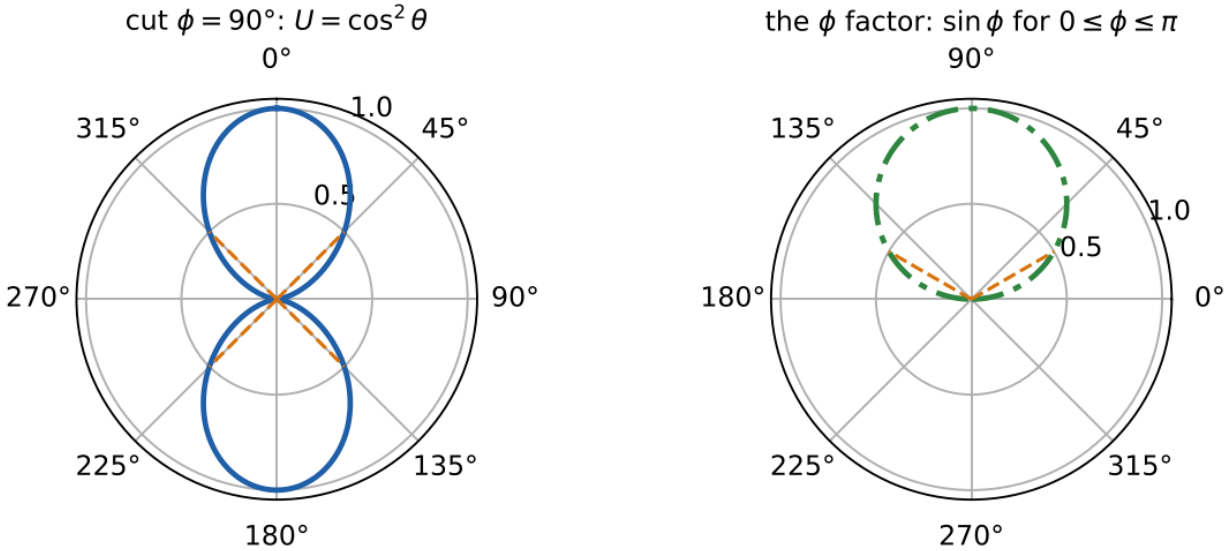


Figure 2: The pattern in two cuts

Reading the figure. Left: the cut $\phi = 90°$, where $U = \cos^2\theta$. It has two equal lobes, along $\theta = 0°$ and $\theta = 180°$; the dashed orange lines reach the half-power points at 45° from each peak. Right: the factor $\sin\phi$, which exists only for $0 \le \phi \le \pi$ and falls to half at 30° and 150°.

Step 1: Half-power beamwidth

In the plane $\phi = 90°$, $U = \cos^2\theta$. Half power means

$$\cos^2\theta = \frac{1}{2} \;\Rightarrow\; \cos\theta = \pm\frac{1}{\sqrt{2}} \;\Rightarrow\; \theta = 45° \text{ or } 135°$$

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The lobe at $\theta = 0$ falls to half at 45° on each side, so

$$\boxed{\text{HPBW} = 2 \times 45° = 90°}$$

The factor $\sin\phi$ falls to half at $\phi = 30°$ and 150°, a width of 120° in ϕ.

Step 2: Beamwidth between first nulls

The first null of $\cos^2\theta$ is at $\theta = 90°$ on each side of the peak:

$$\boxed{\text{FNBW} = 2 \times 90° = 180°}$$

Step 3: Radiated power

$$P_{rad} = \int_0^{\pi}\int_0^{\pi} U \sin\theta \, d\theta \, d\phi = \int_0^{\pi} \sin\phi \, d\phi \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta$$

The first integral is $[-\cos\phi]_0^{\pi} = 2$. For the second, put $u = \cos\theta$, so $du = -\sin\theta \, d\theta$:

$$\int_0^{\pi} \cos^2\theta \sin\theta \, d\theta = \int_{-1}^{1} u^2 \, du = \left[\frac{u^3}{3}\right]_{-1}^{1} = \frac{2}{3}$$

$$P_{rad} = 2 \times \frac{2}{3} = \frac{4}{3}$$

$$\boxed{P_{rad} = 1.33 \text{ W} \quad (\text{for } U_m = 1 \text{ W/sr})}$$

So the beam area is $\Omega_A = 4/3$ sr.

Step 4: Directivity

$$D = \frac{4\pi U_m}{P_{rad}} = \frac{4\pi}{4/3} = 3\pi$$

$$\boxed{D = 3\pi = 9.42 = 9.74 \text{ dBi}}$$

Final Answer: HPBW = 90°, FNBW = 180° (in the plane $\phi = 90°$); $P_{rad} = 4/3$ W for $U_m = 1$ W/sr; $D = 9.42$ (9.74 dBi).

3. Notes and Remarks

- The pattern is awkward at the poles: on the axis $\theta = 0$, every ϕ is the same direction, yet $\sin\phi$ takes different values. This is a feature of the textbook-style formula, not of a real antenna, so the answer reads the peak in the plane $\phi = 90°$.

1.(c) Describe the field regions surrounding an antenna. Marks: 5

1. Theory

1. Three kinds of terms (Core Idea)

The simplest antenna is a **Hertzian dipole**: a very short wire of length dl carrying a current I. Solving Maxwell's equations for it (from its vector potential $A_z = \frac{\mu_0 I\, dl}{4\pi r} e^{-j\beta r}$, as in 4(a) and 4(b) of the 2020–21 paper) gives

$$H_\phi = \frac{I\, dl \, \sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}, \qquad E_\theta = \frac{\eta\, I\, dl \, \sin\theta}{4\pi}\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}$$

(and a radial E_r with only $1/r^2$ and $1/r^3$ terms). Each bracket mixes three kinds of term:

- $1/r^3$, the **electrostatic** term: the field of the charges piling up at the two ends of the wire, like a capacitor.
- $1/r^2$, the **induction** term: the field of the current itself, like an inductor.

- $1/r$, the **radiation** term: the only one that carries power away for good.

Factor out β/r from the bracket of E_θ: it becomes $\left[j + \frac{1}{\beta r} - \frac{j}{(\beta r)^2}\right]$. The three parts have the same size when $\beta r = 1$, i.e. $r = \lambda/2\pi$. Closer in, the $1/r^3$ and $1/r^2$ parts are much larger: they hold energy that flows out and back each cycle. Farther out, only the $1/r$ part remains.

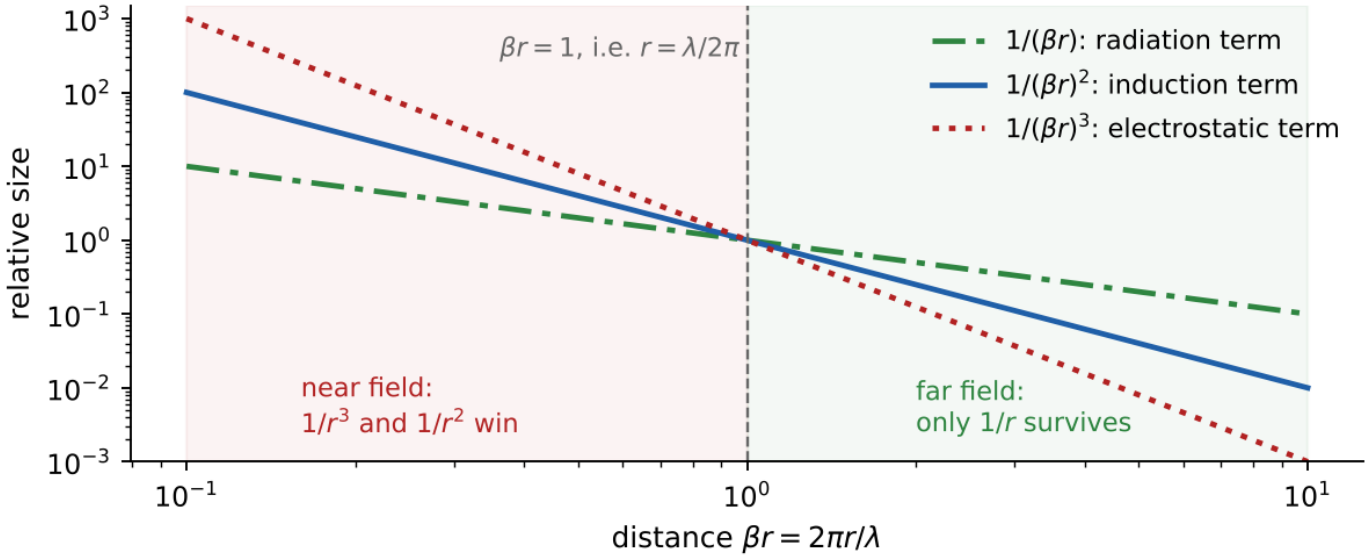


Figure 3: How the three field terms change with distance

Reading the figure. On a log scale, the dotted red $1/r^3$ term and the solid blue $1/r^2$ term are largest close in; the dash-dot green $1/r$ (radiation) term wins beyond $\beta r = 1$.

2. Answer

Near field: the region close to the antenna where the reactive (stored-energy) fields, falling as $1/r^2$ and $1/r^3$, are important. Its pattern changes with distance, $E/H \neq \eta$, and E and H are not in phase. It has two parts:

- **Reactive near field:** $r < 0.62\sqrt{D^3/\lambda}$ (for an antenna small compared with λ, $r < \lambda/2\pi$). Stored energy dominates.
- **Radiating near field (Fresnel region):** $0.62\sqrt{D^3/\lambda} \leq r < 2D^2/\lambda$. Radiation dominates, but the angular pattern still depends on r.

Far field (Fraunhofer region): $r \geq \dfrac{2D^2}{\lambda}$ (and $r \gg D$, $r \gg \lambda$). Only the $1/r$ radiation fields remain: the angular pattern no longer depends on distance, $\mathbf{E} \perp \mathbf{H} \perp \hat{r}$, $E/H = \eta$, and the power density falls as $1/r^2$, like a plane wave locally. Antenna patterns and gain are defined and measured here.

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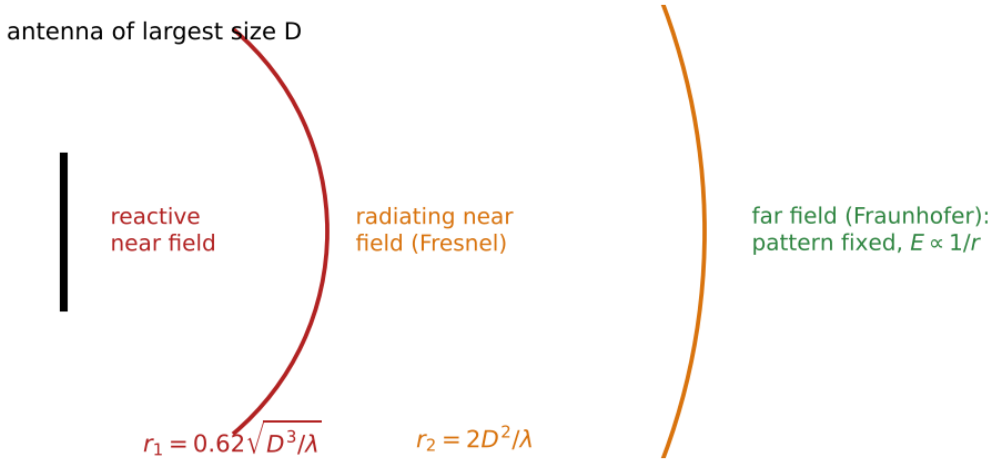


Figure 4: Field regions around an antenna

Reading the figure. The antenna (black bar, largest size D) is at the left. The inner solid red arc at $r_1 = 0.62\sqrt{D^3/\lambda}$ closes the reactive near field; the outer solid orange arc at $r_2 = 2D^2/\lambda$ closes the radiating near field; beyond it lies the far field.

Example

a 3 m dish at 10 GHz ($\lambda = 3$ cm): $r_1 = 0.62\sqrt{27/0.03} = 18.6$ m and $r_2 = 2 \times 9/0.03 = 600$ m; its pattern must be measured more than 600 m away.

	Near field	Far field
Distance	$r < 2D^2/\lambda$	$r \geq 2D^2/\lambda$
Dominant terms	$1/r^3$, $1/r^2$ (stored energy)	$1/r$ (radiation)
Pattern	changes with r	independent of r
E/H	not η, out of phase	η, in phase
Power density	not simply $1/r^2$	$\propto 1/r^2$

Table 1: Near field and far field

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1.(d) For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity. Marks: 8

1. Theory

1. What is an elemental magnetic dipole? (Core Idea)

It is a very small loop of wire carrying a uniform current I. "Very small" means its size is much less than a wavelength, so the current has the same value and phase all round the loop. A bar magnet has a north and a south pole; a small current loop behaves in the same way. Its strength is the **magnetic moment**

$$m = IS$$

where S is the loop's area, and m points along the loop's axis.

2. How do we find fields from a current?

We do not start with **E** and **H** directly, because the integrals are hard. We first find the **magnetic vector potential A**, a helper quantity defined by $\mathbf{B} = \nabla \times \mathbf{A}$. Each piece of current $I\,d\mathbf{l}'$ adds to **A** a term that falls as $1/R$ with distance R and is delayed by the travel time, which gives the factor e^{-jkR}:

$$\mathbf{A} = \frac{\mu}{4\pi} \oint I \, \frac{e^{-jkR}}{R} \, d\mathbf{l}'$$

Here $k = \omega\sqrt{\mu\epsilon} = 2\pi/\lambda$ is the wave number (the same as β).

3. Then how do we get the fields?

Two of Maxwell's equations do it. Outside the source, $\mathbf{H} = \frac{1}{\mu}\nabla \times \mathbf{A}$, and from $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$,

$$\mathbf{E} = \frac{1}{j\omega\epsilon}\nabla \times \mathbf{H}$$

4. Near and far

A field term in $1/r^3$ is large close to the source and dies quickly. It stores energy. A term in $1/r$ falls slowly, and it is the one that carries power to infinity. The answer shows both kinds.

2. Answer

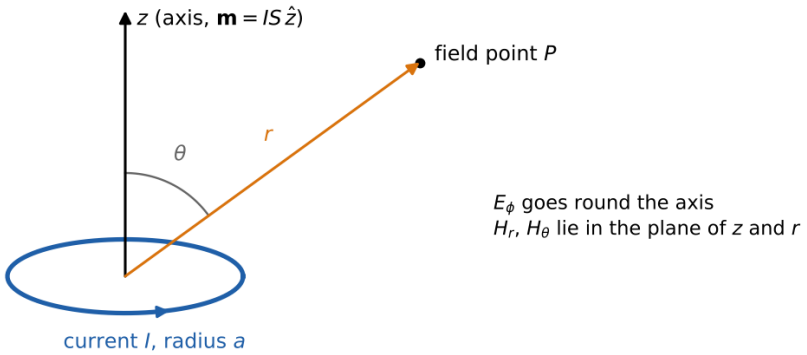


Figure 5: A small current loop on the z axis and a field point P

Reading the figure. The loop lies in the xy plane with its centre at the origin. The field point P is at distance r and at angle θ from the axis. The loop looks the same from every ϕ, so nothing depends on ϕ.

Set-up. Loop of area S and uniform current I, moment $m = IS$, in a medium of μ, ϵ. Let $g(r) = e^{-jkr}/r$, and let $\eta = \sqrt{\mu/\epsilon}$.

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Step 1: Vector potential

With $g(R)$ as above, the potential at P is

$$\mathbf{A} = \frac{\mu I}{4\pi} \oint g(R)\, d\mathbf{l}'$$

Stokes' theorem turns a loop integral of a scalar into a surface integral, $\oint g\, d\mathbf{l}' = \int \hat{n} \times \nabla' g\, dS'$. The loop is tiny, so $\nabla' g$ is the same all over it, and $\hat{n} = \hat{z}$:

$$\oint g\, d\mathbf{l}' = S\, \hat{z} \times \nabla' g$$

Since $R = |\mathbf{r} - \mathbf{r}'|$, a derivative with respect to the source point is minus the derivative with respect to the field point: $\nabla' g = -\nabla g$. For $g(r)$,

$$\nabla g = \frac{dg}{dr}\hat{r} = -\left(jk + \frac{1}{r}\right)\frac{e^{-jkr}}{r}\hat{r}$$

Also $\hat{z} \times \hat{r} = \sin\theta\, \hat{\phi}$. Putting these together:

$$\mathbf{A} = \frac{\mu I}{4\pi} S\, \hat{z} \times (-\nabla g) = \frac{\mu m}{4\pi} \sin\theta \left(\frac{jk}{r} + \frac{1}{r^2}\right) e^{-jkr}\, \hat{\phi}$$

So $\mathbf{A}$ has only a ϕ component. Write $C = m/4\pi$:

$$\frac{A_\phi}{\mu} = C \sin\theta \left(\frac{jk}{r} + \frac{1}{r^2}\right) e^{-jkr}$$

Step 2: The curl for the magnetic field

For a vector with only a ϕ component and no ϕ-dependence, the curl has two components:

$$H_r = \frac{1}{\mu r \sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, A_\phi), \qquad H_\theta = -\frac{1}{\mu r}\frac{\partial}{\partial r}(r A_\phi)$$

Radial part. $\sin\theta\, A_\phi$ contains $\sin^2\theta$, whose derivative is $2\sin\theta\cos\theta$. The $\sin\theta$ cancels with the one below:

$$H_r = 2C\cos\theta \left(\frac{jk}{r^2} + \frac{1}{r^3}\right) e^{-jkr}$$

Polar part. First,

$$\frac{rA_\phi}{\mu} = C\sin\theta \left(jk + \frac{1}{r}\right) e^{-jkr}$$

Differentiate with the product rule, using $\frac{d}{dr}e^{-jkr} = -jke^{-jkr}$:

$$\frac{\partial}{\partial r}\left(\frac{rA_\phi}{\mu}\right) = C\sin\theta \left[-\frac{1}{r^2} - jk\left(jk + \frac{1}{r}\right)\right] e^{-jkr} = C\sin\theta \left[k^2 - \frac{jk}{r} - \frac{1}{r^2}\right] e^{-jkr}$$

Multiply by $-1/r$:

$$H_\theta = C\sin\theta \left(-\frac{k^2}{r} + \frac{jk}{r^2} + \frac{1}{r^3}\right) e^{-jkr}$$

Step 3: The electric field

$\mathbf{H}$ has r and θ components with no ϕ-dependence, so its curl has only a ϕ component:

$$E_\phi = \frac{1}{j\omega\epsilon} \cdot \frac{1}{r}\left[\frac{\partial}{\partial r}(rH_\theta) - \frac{\partial H_r}{\partial\theta}\right]$$

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First derivative. $rH_\theta = C\sin\theta\left(-k^2 + \frac{jk}{r} + \frac{1}{r^2}\right)e^{-jkr}$. The product rule gives

$$\frac{\partial}{\partial r}(rH_\theta) = C\sin\theta\left[-\frac{jk}{r^2} - \frac{2}{r^3} - jk\left(-k^2 + \frac{jk}{r} + \frac{1}{r^2}\right)\right]e^{-jkr}$$

$$= C\sin\theta\left[jk^3 + \frac{k^2}{r} - \frac{2jk}{r^2} - \frac{2}{r^3}\right]e^{-jkr}$$

Second derivative. Only $\cos\theta$ depends on θ:

$$\frac{\partial H_r}{\partial\theta} = -2C\sin\theta\left(\frac{jk}{r^2} + \frac{1}{r^3}\right)e^{-jkr}$$

Subtract. The $1/r^2$ and $1/r^3$ terms cancel:

$$\frac{\partial}{\partial r}(rH_\theta) - \frac{\partial H_r}{\partial\theta} = C\sin\theta\left(jk^3 + \frac{k^2}{r}\right)e^{-jkr}$$

Divide. Since $k/(\omega\epsilon) = \eta$, we have $\dfrac{1}{j\omega\epsilon} = \dfrac{\eta}{jk}$. Dividing by r and multiplying by this:

$$E_\phi = \frac{\eta}{jk}\cdot\frac{C\sin\theta}{r}\left(jk^3 + \frac{k^2}{r}\right)e^{-jkr} = \eta C\sin\theta\left(\frac{k^2}{r} + \frac{k}{jr^2}\right)e^{-jkr}$$

Step 4: The results

Put $C = m/4\pi = IS/4\pi$, and take out k^2/r or the like, so that the brackets are in powers of $1/kr$:

$$\boxed{E_\phi = \frac{\eta k^2 IS}{4\pi r}\sin\theta\left[1 + \frac{1}{jkr}\right]e^{-jkr}}$$

$$\boxed{H_r = \frac{jkIS}{2\pi r^2}\cos\theta\left[1 + \frac{1}{jkr}\right]e^{-jkr}}$$

$$\boxed{H_\theta = -\frac{k^2 IS}{4\pi r}\sin\theta\left[1 + \frac{1}{jkr} - \frac{1}{(kr)^2}\right]e^{-jkr}}$$

and $E_r = E_\theta = H_\phi = 0$.

Step 5: The far field

For $kr \gg 1$ the brackets become 1, and H_r (which goes as $1/r^2$) dies away:

$$E_\phi \approx \frac{\eta k^2 IS}{4\pi r}\sin\theta\, e^{-jkr}, \qquad H_\theta \approx -\frac{E_\phi}{\eta}$$

The ratio is η, as in a plane wave.

Steps to remember

Vector potential → The curl for the magnetic field → The electric field → The results → The far field.

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3. Keyword Explanation

1. Duality

↩ back

The loop's fields look like those of a short electric dipole with **E** and **H** exchanged. This is a quick check of the result.

2. Magnetic moment

↩ back

$m = IS$, in A m^2. A loop of N turns has $m = NIS$.

4. Notes and Remarks

- For a circular loop of radius a, $S = \pi a^2$, so $E_\phi = \eta k^2 a^2 I \sin\theta/(4r)$ in the far field.
- The pattern is $\sin\theta$, zero along the axis and largest in the plane of the loop, the same doughnut as a short electric dipole, but with **E** and **H** swapped.
- **Near the loop**, with $kr \ll 1$: $H_r \approx 2m\cos\theta/(4\pi r^3)$ and $H_\theta \approx m\sin\theta/(4\pi r^3)$, the static field of a magnet.

5. Final Concise Answer

(a) The effective aperture is the area $A_e = P_r/S$ that collects the received power. $D = 4\pi A_e/\lambda^2$, so a larger aperture, in wavelengths, gives a higher directivity and a narrower beam.

(b) HPBW $= 90°$, FNBW $= 180°$, $P_{rad} = 4/3$ W for $U_m = 1$ W/sr, $D = 9.42$ (9.74 dBi).

(c) Reactive near field out to $0.62\sqrt{D^3/\lambda}$, radiating near (Fresnel) field out to $2D^2/\lambda$, then far (Fraunhofer) field with a fixed pattern.

(d) The fields of a small loop of area S and current I are

$$E_\phi = \frac{\eta k^2 IS}{4\pi r}\sin\theta\left[1+\frac{1}{jkr}\right]e^{-jkr}$$

$$H_r = \frac{jkIS}{2\pi r^2}\cos\theta\left[1+\frac{1}{jkr}\right]e^{-jkr}$$

$$H_\theta = -\frac{k^2 IS}{4\pi r}\sin\theta\left[1+\frac{1}{jkr}-\frac{1}{(kr)^2}\right]e^{-jkr}$$

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2. (a) Describe how passive remote sensing is used to detect object. Marks: 5
   (b) Explain how the RADAR range equation is used to detect objects. Marks: 6
   (c) Show the signal to noise ratio of a radio link is related to bandwidth and system temperature. Marks: 5
   (d) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. Marks: 9

2.(a) Describe how passive remote sensing is used to detect object. Marks: 5

1. Theory

1. Everything glows (Core Idea)

Any object warmer than absolute zero gives off electromagnetic noise, because its charges jiggle randomly. At microwave frequencies the power it sends is proportional to its temperature. A **radiometer** is a very sensitive receiver that measures this noise power. It sends nothing itself: that is why the method is **passive**.

2. Brightness temperature

A perfect absorber (a black body) at temperature T emits the most. A real surface emits only a fraction e (its **emissivity**, $0 \le e \le 1$) and reflects the rest of what falls on it. So it "looks" as warm as

$$T_B = e\,T_{phys} \quad \text{(plus the reflected sky)}$$

This is its **brightness temperature**. Soil has $e \approx 0.9$, so it looks warm (about 260 K). Calm water has $e \approx 0.4$ and looks cooler. Metal has $e \approx 0$: it emits almost nothing and mirrors the cold sky (a few kelvin), so it looks very cold.

2. Answer

Passive remote sensing detects and maps objects by measuring the natural thermal (noise) radiation they emit, with a radiometer, without sending any signal.

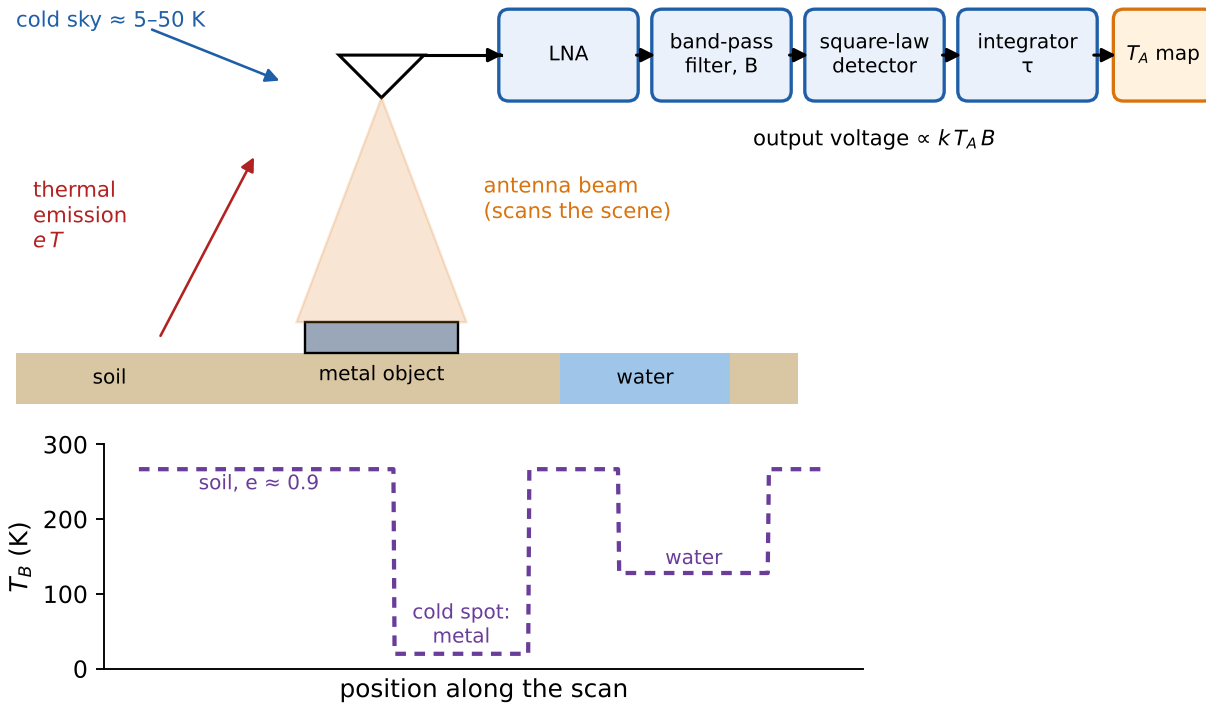


Figure 6: Passive remote sensing with a microwave radiometer

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Reading the figure. The antenna beam (orange shading) looks down at soil, a metal object and water; the soil emits strongly (red arrow), the metal reflects the cold sky (blue arrow). The receiver chain amplifies, filters, detects and averages the noise. The plot below, in purple short dashes, is the brightness temperature along the scan: the metal object shows as a deep cold spot.

How an object is detected:

1. **Emission:** each part of the scene emits noise with brightness temperature $T_B = e\,T_{phys}$; objects with a different emissivity or temperature have a different T_B.
2. **Reception:** the antenna collects it. Its output is the **antenna temperature**, the T_B of the scene averaged over the beam:
$$T_A = \frac{1}{\Omega_A} \iint T_B(\theta,\phi)\, P_n(\theta,\phi)\, d\Omega$$
3. **Measurement:** the radiometer (LNA, band-pass filter of bandwidth B, square-law detector, integrator over a time τ) gives a voltage proportional to the received power $P = k\,T_A\,B$.
4. **Detection:** as the beam scans, an object appears as a change (contrast) ΔT_B against its background. A metal vehicle or ship looks cold; oil on water, ice, wet soil and fires each have their own T_B.
5. **Sensitivity:** the smallest change it can see is $\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$. With $T_{sys} = 500$ K, $B = 100$ MHz and $\tau = 10$ ms, $\Delta T_{min} = 0.5$ K.

Uses: soil moisture, sea ice and sea temperature from satellites; detecting ships, vehicles and concealed metal objects; weather (water vapour, rain). It works day and night and through cloud, and, sending nothing, it cannot be detected or jammed easily.

3. Keyword Explanation

1. Emissivity

↩ back

How well a surface emits compared with a perfect black body at the same temperature; $e = 1 - |\Gamma|^2$, where Γ is its reflection coefficient.

2. Antenna temperature

↩ back

The temperature of a resistor that would deliver the same noise power to the receiver as the antenna does.

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2.(b) Explain how the RADAR range equation is used to detect objects. Marks: 6

1. Theory

1. How radar sees (Core Idea)

A radar sends out a short, powerful pulse. When it hits an object, a tiny part bounces back. The radar listens for this echo: the time it takes gives the distance ($R = ct/2$), and the direction of the antenna gives the bearing. The **radar range equation** tells how strong the echo is, and so how far away an object can still be detected.

2. The spreading twice

The pulse spreads out over a sphere on its way to the target (power density $\propto 1/R^2$), and the echo spreads out again on the way back (another $1/R^2$). That is why the echo falls as $1/R^4$, very fast.

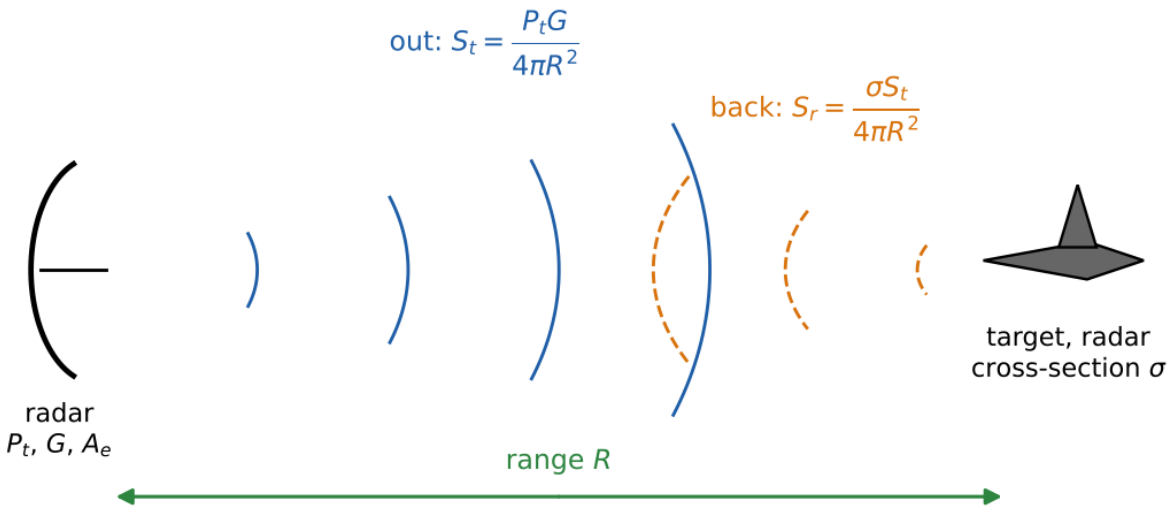


Figure 7: Radar geometry: the outgoing wave and the echo

Reading the figure. The radar on the left sends the outgoing wave (solid blue arcs) towards the target at range R. The target, with radar cross-section σ, sends back an echo (dashed orange arcs) that spreads again on its way back.

2. Answer

Derivation:

1. **Power density at the target:** a transmitter of power P_t with an antenna of gain G gives, at range R,
$$S_t = \frac{P_t G}{4\pi R^2} \ \text{W/m}^2$$
2. **Power scattered by the target:** the target intercepts and re-radiates (isotropically, by definition) the power σS_t, where σ is its **radar cross-section** (m^2).
3. **Power density back at the radar:**
$$S_r = \frac{\sigma S_t}{4\pi R^2} = \frac{P_t G \sigma}{(4\pi)^2 R^4}$$
4. **Received power:** the antenna with effective aperture $A_e = G\lambda^2/4\pi$ collects
$$P_r = S_r A_e = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$$

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Use for detection: an object is detected when the echo is at least the smallest power the receiver can detect, $P_r \geq P_{min}$ (set by the receiver noise, $P_{min} = kT_sB \cdot (S/N)_{min}$). Putting $P_r = P_{min}$ gives the **maximum range**:

$$\boxed{R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}}$$

where:

- P_t = peak transmitted power (W), G = antenna gain, λ = wavelength (m), σ = radar cross-section (m^2), P_{min} = minimum detectable signal (W).

How it is used:

- **Range of the object:** from the echo delay, $R = ct/2$; the equation says whether an echo from that range is strong enough.
- **Design:** it gives the transmitter power, antenna size and receiver sensitivity needed to see a target of a given σ at a given range.
- **Interpretation:** $R_{max} \propto P_t^{1/4}$, so doubling the range needs 16 times the power, or a bigger antenna (G^2), a more sensitive receiver, or pulse integration.
- **Target size:** a large or metallic target (large σ) is seen farther; stealth shapes reduce σ.

3. Keyword Explanation

1. Radar cross-section

↩ back

The area that, intercepting the incident power density and radiating it equally in all directions, would give the same echo as the real target. It depends on the target's size, shape, material and the frequency.

2.(c) Show the signal to noise ratio of a radio link is related to bandwidth and system temperature. Marks: 5

1. Theory

1. What is a radio link? (Core Idea)

A radio link is a transmitter and a receiver with free space between them. The signal that arrives is very weak, and the receiver adds noise of its own. What decides whether the message can be read is the ratio of the two: the **signal-to-noise ratio** (SNR).

2. How much signal arrives?

The transmitter's power spreads over a sphere, so the power density falls as $1/r^2$. A receiving antenna collects the density times its effective area. The result is **Friis' equation**, derived in the Answer.

3. Where does the noise come from?

Everything warm radiates, and electrons in warm conductors move at random. A resistor at temperature T delivers the noise power kTB to a matched load, in a bandwidth B, where $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant. Noise power does not depend on frequency here, only on the width of the band.

4. What is system temperature?

We describe all the noise of a receiver (from the antenna, the feed line and the amplifiers) as if it came from one resistor at a temperature T_{sys}. So the noise power is $N = kT_{sys}B$. A lower T_{sys} means a quieter receiver.

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2. Answer

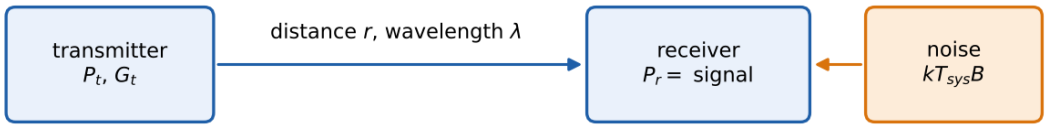


Figure 8: A radio link: the signal arrives and the noise adds at the receiver

Reading the figure. The transmitter sends P_t through an antenna of gain G_t. At the receiver, the signal P_r meets the noise power $kT_{sys}B$.

Step 1: Power density at the receiver

An isotropic antenna spreads P_t evenly over a sphere of area $4\pi r^2$. A gain G_t multiplies the density in the wanted direction:

$$S = \frac{P_t G_t}{4\pi r^2}$$

Step 2: Received power

The receiving antenna has effective aperture $A_{er} = G_r\lambda^2/4\pi$:

$$P_r = SA_{er} = \frac{P_t G_t}{4\pi r^2} \cdot \frac{G_r \lambda^2}{4\pi} = P_t G_t G_r \left(\frac{\lambda}{4\pi r}\right)^2$$

Step 3: Noise power

$$N = kT_{sys}B$$

Step 4: Divide

$$\boxed{\frac{S}{N} = \frac{P_r}{N} = \frac{P_t G_t G_r}{kT_{sys}B}\left(\frac{\lambda}{4\pi r}\right)^2}$$

Steps to remember

Power density at the receiver → Received power → Noise power → Divide.

Interpretation: the SNR is inversely proportional to the bandwidth B and to the system temperature T_{sys}. Doubling either one lowers the SNR by 3 dB.

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2.(d) Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations. Marks: 9

1. Theory

1. Why use many antennas (Core Idea)

One small antenna has a broad pattern. If we put several identical antennas in a line and feed them together, their waves add in some directions and cancel in others. By choosing the spacing and the phase of each feed we can point a narrow beam where we want. A **uniform** linear array has N identical elements, equally spaced by d, fed with equal amplitudes and a constant phase step δ from one element to the next.

2. The path difference

Seen from a far point at angle θ to the array axis, the wave from each element travels $d\cos\theta$ less than the wave from the one before it, a phase lead $\beta d\cos\theta$. Adding the feed phase step, neighbouring elements differ by

$$\psi = \beta d\cos\theta + \delta$$

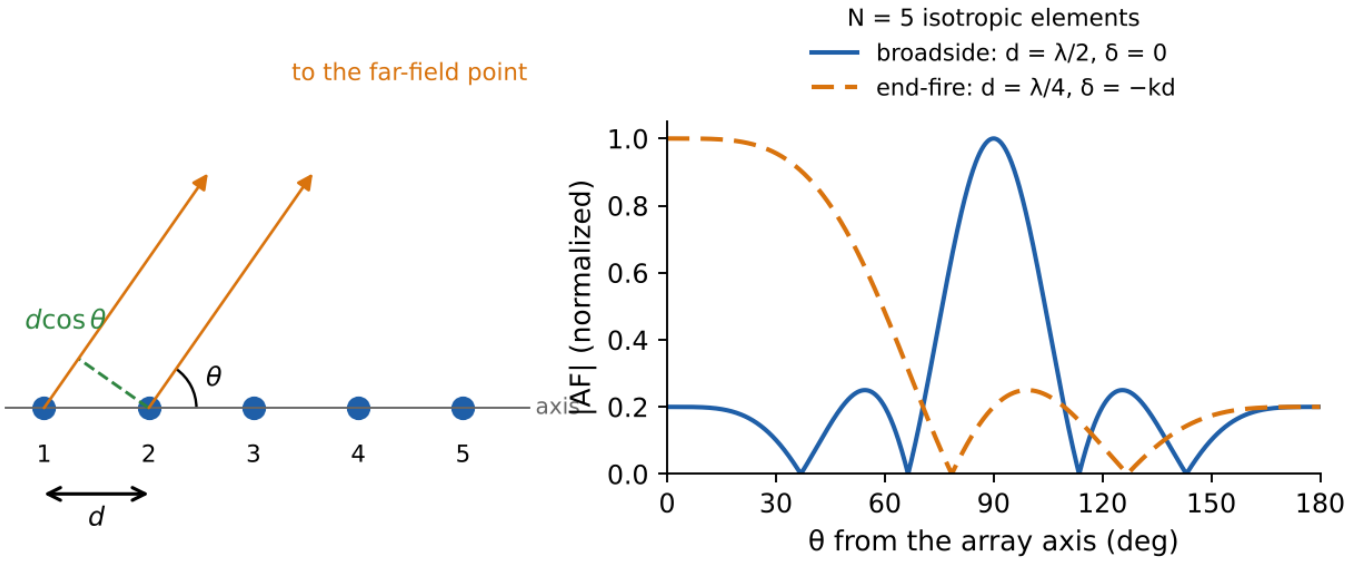


Figure 9: Geometry and array factor of a uniform linear array

Reading the figure. Left: five elements, spacing d, with the extra path $d\cos\theta$ (dashed green) between two neighbours. Right: the array factor of five elements: broadside ($d = \lambda/2$, $\delta = 0$, solid blue) with its main beam at 90°, and end-fire ($d = \lambda/4$, $\delta = -\beta d$, dashed orange) with its beam along the axis at 0°.

2. Answer

Set-up. Take N identical isotropic elements along a line, a spacing d apart, with equal amplitudes. Each element's feed leads the one before it by the phase δ. A far-away point lies in the direction θ from the line, so the rays from all the elements are parallel.

Step 1: Phase difference between neighbours

The ray from each element has a path shorter by $d\cos\theta$ than the ray from the element before it. That shorter path is a phase lead of $\beta d\cos\theta$, and the feed adds δ. So each element leads the one before it by

$$\psi = \beta d\cos\theta + \delta$$

Step 2: Add the elements

With element 1 as the reference, the total field (the **array factor**) is

$$AF = 1 + e^{j\psi} + e^{j2\psi} + \cdots + e^{j(N-1)\psi}$$

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Step 3: Sum the series

This is a geometric series. Call the sum S and let $y = e^{j\psi}$. Then $yS = y + y^2 + \cdots + y^N$. Subtracting, every middle term cancels:

$$S - yS = 1 - y^N \;\Rightarrow\; AF = \frac{1 - e^{jN\psi}}{1 - e^{j\psi}}$$

Step 4: Turn the sum into sines

Take $e^{jN\psi/2}$ out of the top and $e^{j\psi/2}$ out of the bottom, and use $e^{jA} - e^{-jA} = 2j \sin A$:

$$1 - e^{jN\psi} = e^{jN\psi/2}\left(e^{-jN\psi/2} - e^{jN\psi/2}\right) = -2j\, e^{jN\psi/2} \sin\frac{N\psi}{2}$$

$$1 - e^{j\psi} = -2j\, e^{j\psi/2} \sin\frac{\psi}{2}$$

Dividing:

$$AF = e^{j(N-1)\psi/2}\, \frac{\sin(N\psi/2)}{\sin(\psi/2)}$$

Step 5: Normalize

The exponential only sets the phase, which depends on the choice of reference, and has magnitude 1. As $\psi \to 0$, each sine equals its small angle, so the ratio tends to $(N\psi/2)/(\psi/2) = N$. That is the peak value. Dividing by N:

$$\boxed{|AF_n| = \left|\frac{\sin(N\psi/2)}{N \sin(\psi/2)}\right|, \qquad \psi = \beta d \cos\theta + \delta}$$

Steps to remember

Phase difference between neighbours → Add the elements → Sum the series → Turn the sum into sines → Normalize.

Main beam direction: the maximum ($|AF_n| = 1$) is at $\psi = 0$:

$$\beta d \cos\theta_m + \delta = 0 \;\Rightarrow\; \theta_m = \cos^{-1}\left(-\frac{\delta}{\beta d}\right)$$

- **Broadside** ($\theta_m = 90°$): $\cos\theta_m = 0$, so $\delta = 0$.
- **End-fire** ($\theta_m = 0°$): $\cos\theta_m = 1$, so $\delta = -\beta d$.
- Changing δ electronically steers the beam (phased array).

Null locations: a null needs the top to be zero while the bottom is not. So $\sin(N\psi/2) = 0$ with $\psi \neq 0$:

$$\frac{N\psi}{2} = \pm k\pi \;\Rightarrow\; \psi = \pm\frac{2k\pi}{N}, \qquad k = 1, 2, 3, \ldots \quad (k \neq N, 2N, \ldots)$$

Put in $\psi = \beta d \cos\theta_k + \delta$ and solve for θ_k:

$$\theta_k = \cos^{-1}\left[\frac{1}{\beta d}\left(-\delta \pm \frac{2k\pi}{N}\right)\right]$$

For broadside, $\delta = 0$ and $\beta = 2\pi/\lambda$, so $\theta_k = \cos^{-1}\left(\pm\dfrac{k\lambda}{Nd}\right)$.

Width of the main beam, broadside.

First-null beamwidth. The first nulls ($k = 1$) are at $\cos^{-1}(\pm\lambda/Nd)$, on either side of 90°. The beam reaches from 90° down to $\cos^{-1}(\lambda/Nd)$ on each side:

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$$\text{FNBW} = 2\left[\frac{\pi}{2} - \cos^{-1}\frac{\lambda}{Nd}\right] = 2\sin^{-1}\frac{\lambda}{Nd} \approx \frac{2\lambda}{Nd}$$

The last step uses $\sin^{-1} y \approx y$ for $Nd \gg \lambda$.

Half-power beamwidth. Half power means $|AF_n| = 1/\sqrt{2} = 0.707$. Near the main beam ψ is small, so $\sin(\psi/2) \approx \psi/2$ and, with $u = N\psi/2$,

$$|AF_n| \approx \frac{\sin u}{u}$$

Solving $\sin u/u = 0.707$ numerically gives $u = 1.391$. A direction Δ away from broadside has $\theta = 90° - \Delta$, so $\psi = \beta d \sin\Delta \approx \beta d\Delta$. Then

$$\frac{N\beta d\Delta}{2} = 1.391 \;\Rightarrow\; \Delta = \frac{2(1.391)}{N(2\pi/\lambda)d} = 0.443\,\frac{\lambda}{Nd}$$

The beam spreads by Δ on each side:

$$\text{HPBW} = 2\Delta \approx 0.886\,\frac{\lambda}{Nd}\ \text{rad}$$

Width of the main beam, end-fire. With $\delta = -\beta d$, $\psi = \beta d(\cos\theta - 1)$. The first null has $\psi = -2\pi/N$:

$$\cos\theta_1 = 1 - \frac{\lambda}{Nd}$$

The beam is centred on $\theta = 0$ and reaches θ_1 on each side, so $\text{FNBW} = 2\theta_1 = 2\cos^{-1}\left(1 - \frac{\lambda}{Nd}\right)$. For small θ_1, $\cos\theta_1 \approx 1 - \theta_1^2/2$, which gives $\theta_1 \approx \sqrt{2\lambda/Nd}$:

$$\text{FNBW} \approx 2\sqrt{\frac{2\lambda}{Nd}}$$

The beam narrows as the array length Nd grows.

Example

$N = 5$, $d = \lambda/2$, broadside: nulls at $\cos^{-1}(\pm 0.4) = 66.4°, 113.6°$ and $\cos^{-1}(\pm 0.8) = 36.9°, 143.1°$; FNBW $= 47.2°$; HPBW $= 20.4°$; first side lobe 0.25 (-12 dB).

Side lobes and grating lobes: side lobes lie between the nulls, the first about -13 dB for a long array. To avoid extra main beams (grating lobes), $d < \lambda$ for broadside and $d < \lambda/2$ for end-fire.

3. Final Concise Answer

(a) A radiometer's antenna collects the thermal noise that objects emit. An object's brightness temperature $T_B = e\,T_{phys}$ differs from its background, so the map of received noise power $P = kT_AB$ shows it.

(b) $R_{max} = [P_tG^2\lambda^2\sigma/((4\pi)^3P_{min})]^{1/4}$; an object is detected when the echo power is at least P_{min}, and its range is $R = ct/2$.

(c) $S/N = \dfrac{P_tG_tG_r}{kT_{sys}B}\left(\dfrac{\lambda}{4\pi r}\right)^2$: inversely proportional to B and T_{sys}.

(d) $|AF_n| = \left|\dfrac{\sin(N\psi/2)}{N\sin(\psi/2)}\right|$ with $\psi = \beta d\cos\theta + \delta$; main beam at $\psi = 0$; nulls at $\psi = \pm 2k\pi/N$; broadside HPBW $\approx 0.886\lambda/Nd$.

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3. (a) Write short notes on.

   (i) Lorentz reciprocity theorem.
   (ii) Image theory. Marks: 6

   (b) Write the steps to determine line input impedance using Smith Chart. Marks: 4

   (c) A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line. Marks: 7

   (d) A 50 Ω transmission line is connected to a load impedance $35 - j47.5$ Ω. Find the position and length of a short circuited stub required to match the line. Marks: 8

3.(a) Write short notes on. (i) Lorentz reciprocity theorem. (ii) Image theory. Marks: 6

1. Theory

1. What does reciprocity mean? (Core Idea)

Think of two antennas. If we send from antenna 1 and receive on antenna 2, we get some result. Reciprocity says that swapping the roles gives the same result. The path works equally well in both directions.

2. Why does a conductor act like a mirror?

On a perfect conductor the tangential electric field must be zero. Charges in the metal move until that is true. Their field, seen from above the metal, is exactly the field of a mirror copy of the source placed below it. So instead of solving for the charges, we replace the metal by the mirror copy.

2. Answer

i. Lorentz reciprocity theorem

Take two sets of sources, $(\mathbf{J}_1, \mathbf{M}_1)$ and $(\mathbf{J}_2, \mathbf{M}_2)$, at one frequency, in a linear medium. They make the fields $(\mathbf{E}_1, \mathbf{H}_1)$ and $(\mathbf{E}_2, \mathbf{H}_2)$. Then

$$\nabla \cdot (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) = \mathbf{E}_2 \cdot \mathbf{J}_1 - \mathbf{E}_1 \cdot \mathbf{J}_2 + \mathbf{H}_1 \cdot \mathbf{M}_2 - \mathbf{H}_2 \cdot \mathbf{M}_1$$

Without magnetic sources, and with the fields going to zero fast enough at infinity,

$$\boxed{\int \mathbf{E}_2 \cdot \mathbf{J}_1 \, dV = \int \mathbf{E}_1 \cdot \mathbf{J}_2 \, dV}$$

Meaning: the field of source 2 acting on source 1 equals the field of source 1 acting on source 2.
Use:

- An antenna has the same pattern, gain and impedance when transmitting and when receiving, so it can be measured either way.
- A network of linear, isotropic materials has $S_{12} = S_{21}$ and $Z_{12} = Z_{21}$.

Limit: it fails in non-reciprocal media, such as a magnetized ferrite or a plasma in a magnetic field.

ii. Image theory

A source above a large, perfectly conducting plane gives, above the plane, the same field as the source and its image together, with the plane removed. The image is the same distance below the plane.

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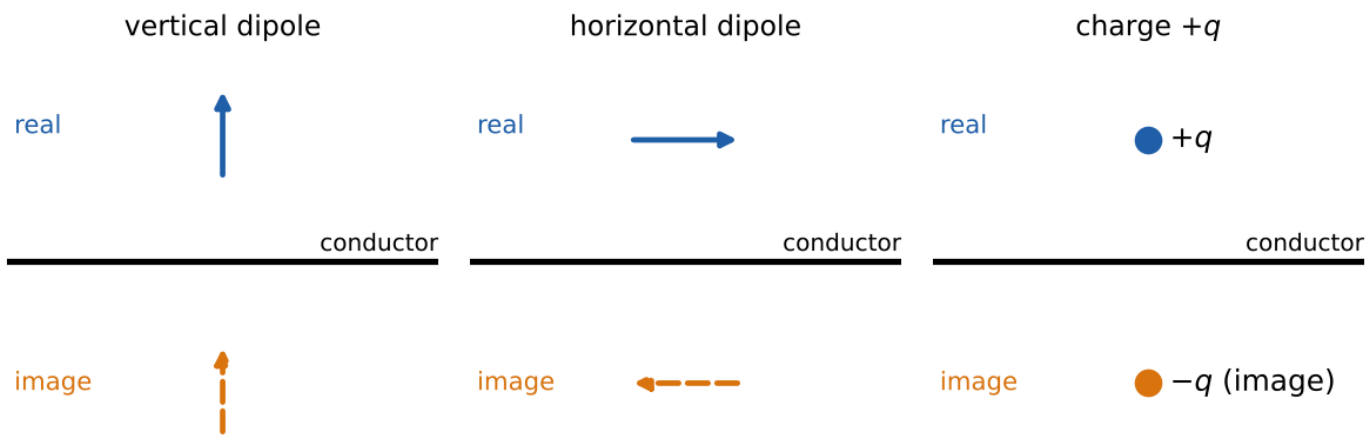


Figure 10: Sources above a conductor and their images

Reading the figure. Blue is the real source and dashed orange its image. A vertical current has an image in the same direction. A horizontal current has an image in the opposite direction. A charge $+q$ has the image $-q$. In each case the tangential **E** of the pair is zero on the plane, which is the boundary condition.

Why it works: the uniqueness theorem says that fields that meet the sources and the boundary conditions in a region are the only solution there.

Use: the monopole above a ground plane is half of a dipole. It has half the radiation resistance and twice the directivity of the dipole, because the power goes only into the upper half space.

Limit: the result holds only above the plane. Below the conductor the real field is zero.

3.(b) Write the steps to determine line input impedance using Smith Chart. Marks: 4

1. Theory

1. What is the input impedance of a line? (Core Idea)

A load Z_L seen through a length of line no longer looks like Z_L. The reflected wave returns with a delay, and it changes the ratio of voltage to current at the input.

2. Where does the formula come from?

Put the load at $z = 0$ and the source at $z = -l$. On a lossless line,

$$V(z) = V_0^+(e^{-j\beta z} + \Gamma_L e^{j\beta z}), \qquad I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma_L e^{j\beta z})$$

where $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0)$. At $z = -l$,

$$Z_{in} = \frac{V(-l)}{I(-l)} = Z_0 \frac{e^{j\beta l} + \Gamma_L e^{-j\beta l}}{e^{j\beta l} - \Gamma_L e^{-j\beta l}}$$

Divide the top and bottom by $e^{j\beta l}$:

$$Z_{in} = Z_0 \frac{1 + \Gamma_L e^{-j2\beta l}}{1 - \Gamma_L e^{-j2\beta l}}$$

Put in Γ_L, multiply the top and bottom by $(Z_L + Z_0)e^{j\beta l}$, and use $e^{\pm j\beta l} = \cos\beta l \pm j \sin\beta l$:

$$Z_{in} = Z_0 \frac{2Z_L \cos\beta l + j2Z_0 \sin\beta l}{2Z_0 \cos\beta l + j2Z_L \sin\beta l}$$

Divide by $2\cos\beta l$:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$$

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3. Why is a Smith chart useful?

The formula above is slow with complex numbers. In the second form, moving along the line only rotates Γ: $\Gamma(l) = \Gamma_L e^{-j2\beta l}$. Its size stays the same. The Smith chart is the plane of Γ, with circles of constant r and x drawn on it, so a move along the line is simply a rotation.

4. How far do we rotate?

One full turn of 360° on the chart is $2\beta l = 2\pi$, which means $l = \lambda/2$. The outer scale is marked in wavelengths, from 0 to 0.5, so the distance can be read directly.

2. Answer

1. **Normalize the load:** $z_L = Z_L/Z_0 = r + jx$.
2. **Plot** z_L where the circle of constant r meets the arc of constant x.
3. **Draw the SWR circle:** a circle centred on the chart's centre through z_L. It is the circle of constant $|\Gamma|$.
4. **Read the starting position:** draw a line from the centre through z_L to the outer scale, and read it on the "wavelengths toward generator" (WTG) scale.
5. **Rotate clockwise** by the line length in wavelengths: new WTG reading = old reading $+l/\lambda$, subtracting 0.5 if it passes 0.5.
6. **Find z_{in}:** draw a line from the new position to the centre. It cuts the SWR circle at z_{in}.
7. **Denormalize:** $Z_{in} = z_{in} Z_0$.

3.(c) A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line. Marks: 7

1. Theory

1. What is reflection? (Core Idea)

A wave reaches the load. If the load is not equal to Z_0, it cannot absorb the wave completely, and part comes back. The **reflection coefficient** is the ratio of the reflected voltage to the incident voltage at the load:

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

This comes from $V = V^+ + V^-$ and $I = (V^+ - V^-)/Z_0$ at the load, with $V/I = Z_L$.

2. What is a standing wave?

The incident and reflected waves add. At some places they are in phase, and the voltage is a maximum $V_{max} = |V^+|(1+|\Gamma|)$. At others they are out of phase, and it is a minimum $V_{min} = |V^+|(1-|\Gamma|)$. These places stay fixed.

3. What is the SWR?

$$S = \frac{V_{max}}{V_{min}} = \frac{1+|\Gamma|}{1-|\Gamma|}$$

It is 1 for a perfect match and infinite for a total reflection.

4. Where is the voltage largest?

At a distance d from the load, the reflected wave has travelled $2d$ further than the incident wave. So the voltage has the form $1 + |\Gamma|e^{j(\theta - 2\beta d)}$, where θ is the angle of Γ_L. It is largest when the phase is a multiple of 2π.

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2. Answer

Given:

- $l = 0.434\lambda$, $Z_0 = 100\ \Omega$, lossless line
- $Z_L = 260 + j180\ \Omega$

Step 1: Reflection coefficient

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{160 + j180}{360 + j180}$$

Multiply the top and bottom by the conjugate of the bottom:

$$\Gamma_L = \frac{(160 + j180)(360 - j180)}{360^2 + 180^2} = \frac{90\,000 + j36\,000}{162\,000} = 0.556 + j0.222$$

$$\boxed{\Gamma_L = 0.598\angle 21.8^\circ}$$

Step 2: Standing wave ratio

$$S = \frac{1 + 0.598}{1 - 0.598} = \frac{1.598}{0.402}$$

$$\boxed{S = 3.98}$$

Step 3: Location of the voltage maximum

The maximum needs $\theta - 2\beta d = -2n\pi$. With $\beta = 2\pi/\lambda$, and $\theta = 21.8^\circ = 0.3805$ rad, the nearest one ($n = 0$) has

$$d_{max} = \frac{\theta}{2\beta} = \frac{\theta\lambda}{4\pi} = \frac{0.3805}{4\pi}\lambda = 0.0303\lambda$$

$$\boxed{d_{max} = 0.0303\lambda \text{ from the load}}$$

The next maximum would be $\lambda/2$ further, at 0.530λ, which is beyond the 0.434λ line. So there is one voltage maximum on this line. The minimum is $\lambda/4$ from it, at 0.280λ.

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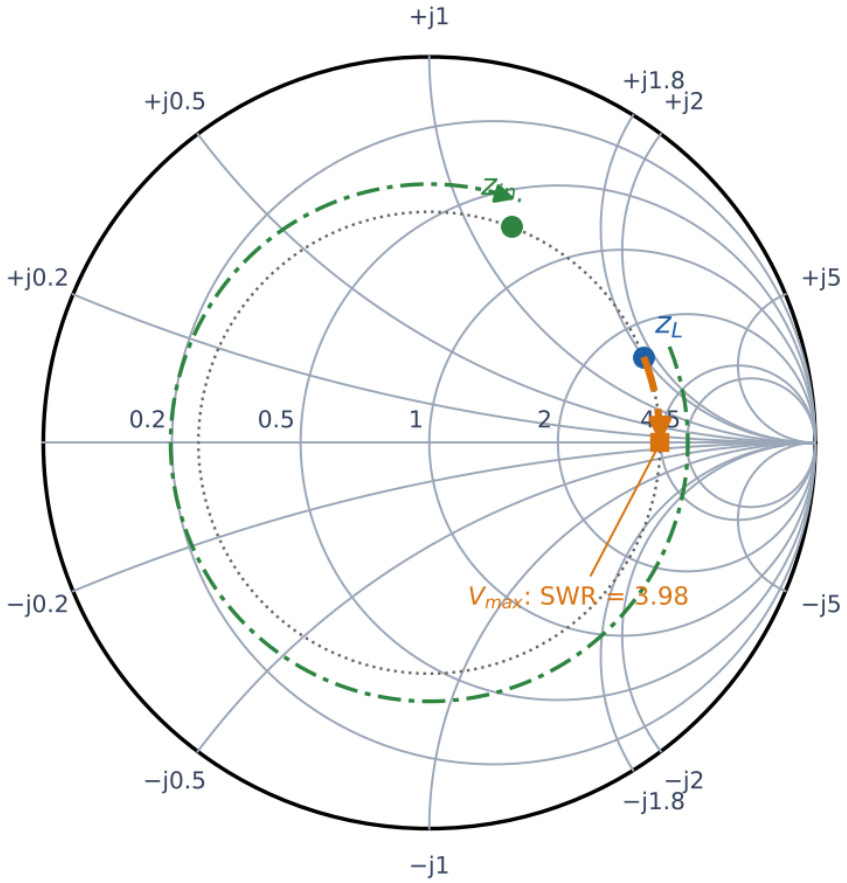


Figure 11: The load on a Smith chart

Reading the figure. The blue dot is $z_L = 2.6 + j1.8$. The dotted circle is the SWR circle, $|\Gamma| = 0.598$. Moving clockwise by 0.0303λ (thick orange) reaches the positive real axis, where $r = 3.98$: this is the voltage maximum, and it reads the SWR directly. The outer green arc goes the whole 0.434λ to the input, at the green dot.

Input impedance (the line length is given, so it is useful to find). With $\tan(2\pi \times 0.434) = -0.440$:

$$Z_{in} = 100\,\frac{(260 + j180) + j100(-0.440)}{100 + j(260 + j180)(-0.440)} = 68.6 + j119.7\ \Omega$$

Final Answer: $\Gamma_L = 0.598\angle 21.8°$; SWR $= 3.98$; first voltage maximum 0.0303λ from the load.

3.(d) A 50 Ω transmission line is connected to a load impedance $35 - j47.5\ \Omega$. Find the position and length of a short circuited stub required to match the line. Marks: 8

1. Theory

1. The idea of a stub (Core Idea)

A mismatched load sends part of the wave back. As we move from the load towards the generator, the admittance seen on the line changes. At some point its real part becomes exactly Y_0: then $y = 1 + jb$. If we connect, at that point and in parallel, a short-circuited piece of line (a **stub**) whose input admittance is $-jb$, the two add to $y = 1$: a perfect match from there to the generator. The stub is chosen short-circuited because a short is easy to make and does not radiate. The same method is used in 2(b) of the 2020–21 paper.

2. Two numbers to find

- the **position** d: distance from the load to the stub, where $\mathrm{Re}(y) = 1$;
- the **length** l: the stub length that gives $-jb$. A short-circuited line of length l has $Z_{sc} = jZ_0 \tan \beta l$ (4(c)), so its normalized admittance is $y_{stub} = \dfrac{Z_0}{jZ_0 \tan \beta l} = -j \cot \beta l$: any pure susceptance can be made by choosing l.

3. Finding the position

At a distance d from the load, with $t = \tan \beta d$, the line shows $Z = Z_0 \dfrac{Z_L + jZ_0 t}{Z_0 + jZ_L t}$. Writing $Z_L = R_L + jX_L$ and taking $Y = 1/Z = G + jB$:

$$Y = \frac{(Z_0 - X_L t) + jR_L t}{Z_0\,[R_L + j(X_L + Z_0 t)]} \;\Rightarrow\; G = \frac{R_L(1+t^2)}{R_L^2 + (X_L + Z_0 t)^2}, \qquad B = \frac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0\,[R_L^2 + (X_L + Z_0 t)^2]}$$

(multiply the top and bottom by the conjugate of the bracket and keep the real and imaginary parts). The match needs $G = Y_0 = 1/Z_0$:

$$Z_0 R_L(1+t^2) = R_L^2 + (X_L + Z_0 t)^2 \;\Rightarrow\; Z_0(R_L - Z_0)\,t^2 - 2X_L Z_0\,t + (Z_0 R_L - R_L^2 - X_L^2) = 0$$

a quadratic in t, whose two roots give the two possible stub positions:

$$t = \frac{X_L \pm \sqrt{R_L\,[(Z_0 - R_L)^2 + X_L^2]\,/Z_0}}{R_L - Z_0}$$

The same steps can be done graphically on the Smith chart (below).

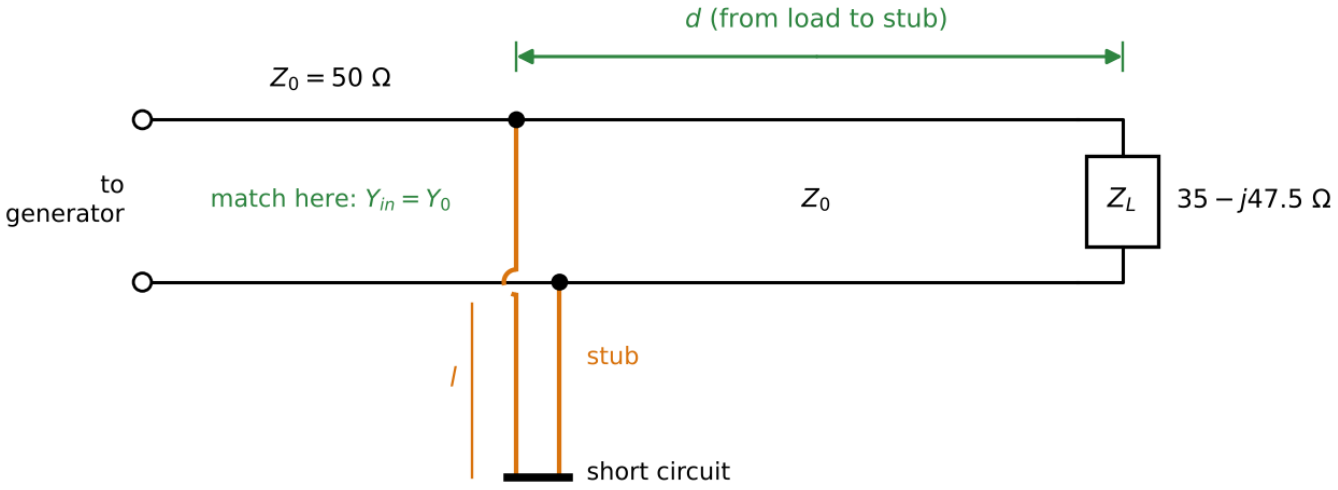


Figure 12: The stub between the load and the generator

Reading the figure. The load $35 - j47.5\ \Omega$ is at the right. At a distance d from it (solid green), the short-circuited stub of length l (orange) hangs in parallel. From the stub to the generator the line is matched.

2. Answer

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Given:

- $Z_0 = 50\ \Omega$, $Z_L = 35 - j47.5\ \Omega$, short-circuited shunt stub of the same Z_0.

Step 1: Normalize

$$z_L = \frac{35 - j47.5}{50} = 0.70 - j0.95, \qquad y_L = \frac{1}{z_L} = 0.503 + j0.682$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.512\angle -78.33°, \qquad S = \frac{1 + 0.512}{1 - 0.512} = 3.09$$

Step 2: Position of the stub

With $t = \tan \beta d$, the condition $G = Y_0$ (Theory, part 3) gives

$$t = \frac{X_L \pm \sqrt{R_L\left[(Z_0 - R_L)^2 + X_L^2\right]/Z_0}}{R_L - Z_0} = \frac{-47.5 \pm 41.68}{-15}$$

$$t_1 = 0.3883, \qquad t_2 = 5.945$$

$$d_1 = \frac{1}{2\pi}\tan^{-1} t_1 = 0.0589\lambda, \qquad d_2 = \frac{1}{2\pi}\tan^{-1} t_2 = 0.2235\lambda$$

At these points $y = 1 + j1.191$ and $y = 1 - j1.191$.

Step 3: Length of the stub

The line susceptance there is $B = \pm 1.191\, Y_0$ (from the formula for B above). The stub must cancel it: $-j \cot \beta l = \mp j 1.191$.

$$l_1 = \frac{1}{2\pi}\tan^{-1}\frac{1}{1.191} = 0.1112\lambda, \qquad l_2 = \frac{1}{2\pi}\left[\pi - \tan^{-1}\frac{1}{1.191}\right] = 0.3888\lambda$$

$$\boxed{d_1 = 0.059\lambda,\ l_1 = 0.111\lambda \quad \text{or} \quad d_2 = 0.224\lambda,\ l_2 = 0.389\lambda}$$

On the Smith chart:

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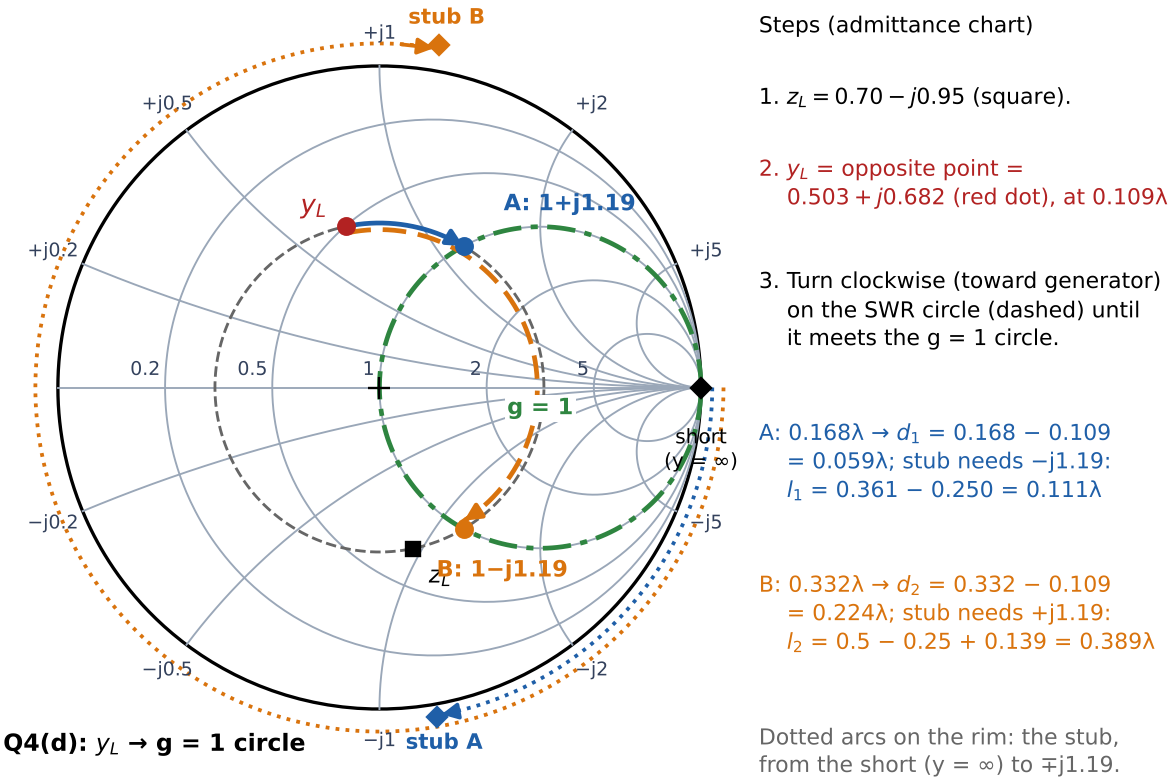


Figure 13: Single-stub matching on the Smith chart

Reading the figure. The square is z_L and the red dot y_L, opposite it on the dashed SWR circle (at 0.109λ on the WTG scale). Turning clockwise meets the dash-dot green $g = 1$ circle at A ($1 + j1.19$, solid blue arc, 0.059λ) and B ($1 - j1.19$, orange arc, 0.224λ). The dotted arcs on the rim are the stubs, from the short ($y = \infty$) to $\mp j1.19$.

Final Answer: the stub is placed 0.059λ from the load with a length of 0.111λ (solution A, shorter and preferred), or 0.224λ from the load with a length of 0.389λ (solution B).

3. Notes and Remarks

- **Check:** at d_1 the line admittance is $1 + j1.1907$ and the stub gives $-j1.1907$: the sum is exactly 1.
- Solution A is preferred: the stub is close to the load, so the stretch of line carrying the high SWR (3.09) is short, giving a wider bandwidth and less loss.

4. Final Concise Answer

(a) Reciprocity: $\int \mathbf{E}_2 \cdot \mathbf{J}_1 \, dV = \int \mathbf{E}_1 \cdot \mathbf{J}_2 \, dV$, so an antenna behaves the same transmitting and receiving. Image theory: a conductor can be replaced by the mirror image of the source.

(b) Normalize, plot z_L, draw the SWR circle, rotate clockwise by l/λ towards the generator, read z_{in}, multiply by Z_0.

(c) $\Gamma_L = 0.598\angle 21.8°$, SWR = 3.98, first voltage maximum 0.0303λ from the load.

(d) Stub at 0.059λ from the load with length 0.111λ, or at 0.224λ with length 0.389λ.

4. (a) Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure [*sic*: should be curve] of the resistor. Marks: 6
   (b) What is microstrip antenna? Mention some of the applications of microstrip line. Marks: 6
   (c) Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length. Marks: 7
   (d) How is quarter wavelength transmission line used for impedance matching? Marks: 6

4.(a) ⚠ Note on the paper: "cure" is a misprint for "curve", so the curve of impedance against frequency is drawn.

1. Theory

1. Why does a resistor change at high frequency? (Core Idea)

At low frequency a resistor is just R. At RF, its other small parasitic properties start to matter. Its leads are short wires, and any wire has inductance. Its two ends are conductors close together, and those make a small capacitor.

2. How do L and C behave?

An inductor has $X_L = \omega L$, which grows with frequency. A capacitor has $X_C = 1/\omega C$, which falls. At high enough frequency even a few nanohenries or picofarads matter more than R.

2. Answer

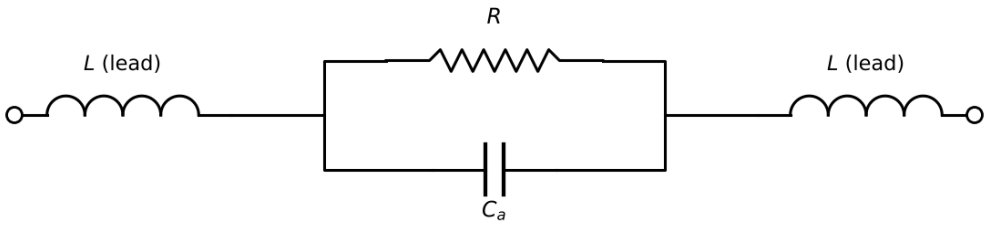


Figure 14: High-frequency equivalent circuit of a resistor

Equivalent circuit. R is the wanted resistance. C_a is the charge-separation capacitance between the ends, in parallel with R. L is the inductance of each lead, in series. A leaded resistor also has a small capacitance between its leads; it is left out here because C_a dominates.

Impedance:

$$Z = j\omega(2L) + \frac{1}{\frac{1}{R} + j\omega C_a}$$

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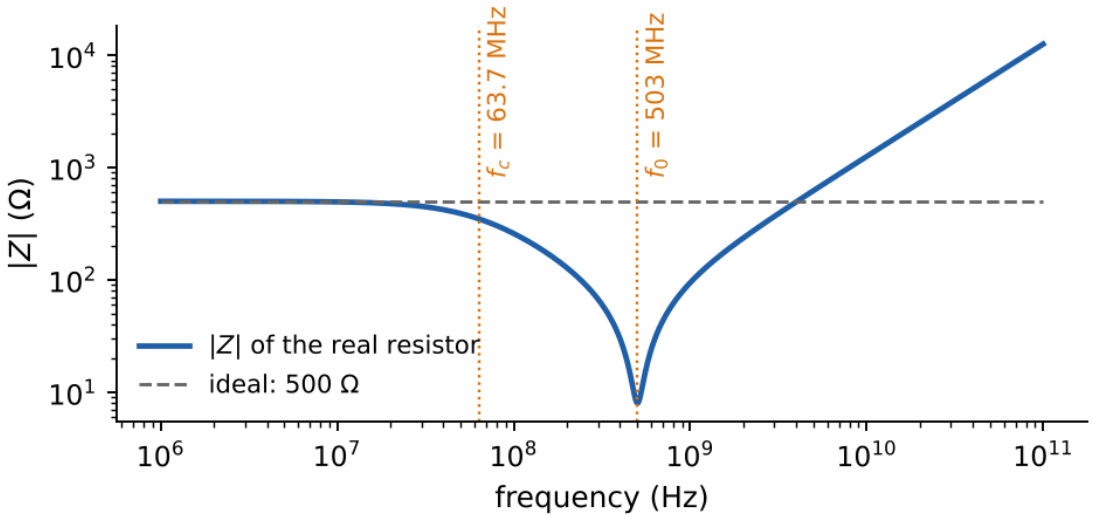


Figure 15: Typical frequency response of a 500 ohm resistor

Reading the figure. The curve is drawn for illustrative values $R = 500\ \Omega$, $C_a = 5$ pF and $2L = 20$ nH; the paper gives none. The dashed grey line is the ideal resistor.

- **Below** $f_c = 1/(2\pi RC_a) = 63.7$ MHz: $|Z| \approx R$.
- **Above** f_c: C_a takes over and $|Z|$ falls.
- **At** $f_0 = 1/(2\pi\sqrt{2LC_a}) = 503$ MHz: the capacitor and the leads resonate in series. $|Z|$ has its minimum, 7.9 Ω.
- **Above** f_0: the leads dominate and $|Z| \approx 2\pi f(2L)$ rises by 20 dB per decade.

3. Keyword Explanation

1. Parasitic

↩ back

A small unwanted element that comes with a real component.

4.(b) What is microstrip antenna? Mention some of the applications of microstrip line. Marks: 6

1. Theory

1. What is a microstrip line? (Core Idea)

It is a thin metal strip on one side of a dielectric board, with a metal ground plane on the other side. The wave travels partly in the board and partly in the air above. It is how transmission lines are made on printed circuit boards.

2. How can a line become an antenna?

A line with its fields held tightly in the substrate radiates very little. If the strip is made wide and about half a wavelength long, it becomes a resonator, and the fringing fields at its two open ends radiate.

2. Answer

Microstrip antenna (patch antenna): a metal patch on a thin dielectric substrate over a ground plane, about half a wavelength long in the substrate. It radiates from the fringing fields at its open edges, mostly broadside (perpendicular to the board).

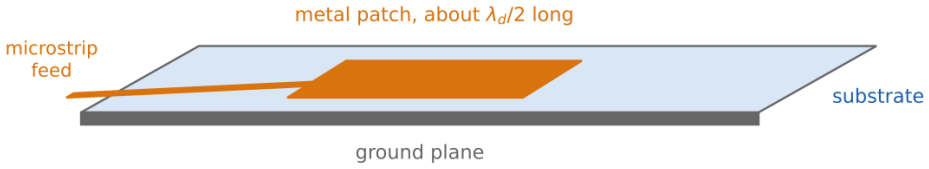


Figure 16: A microstrip patch fed by a microstrip line

Reading the figure. The orange patch sits on the blue substrate above the grey ground plane, and a thin microstrip line feeds it.

Properties: low profile, light, cheap, and easy to make in arrays. Its weaknesses are a narrow bandwidth, low power handling and modest gain.

Applications of microstrip line:

- Microwave integrated circuits (MICs and MMICs), connecting devices on a chip or board.
- Matching networks, stubs and quarter-wave transformers.
- Filters, couplers and power dividers.
- Feed networks of patch arrays.
- Mobile phones, GPS, radar and satellite receivers.

Applications of microstrip antennas: mobile phones, GPS, aircraft and missiles, Wi-Fi and RFID.

4.(c) Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length. Marks: 7

1. Theory

1. What happens at a short? (Core Idea)

A short circuit forces the voltage at the end of the line to zero. The incident wave cannot be absorbed, so all of it comes back, with its sign reversed so that the two voltages cancel at the short. The incident and reflected waves form a pure standing wave.

2. Why does this matter?

A shorted line stores energy without losing it, so its input is a pure reactance. Choosing the length gives any inductance or capacitance. This is how a stub works.

2. Answer

Set-up. A lossless line with the short at $z = 0$, so $Z_L = 0$.

Step 1: General solution

$$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \qquad I(z) = \frac{1}{Z_0}\left(V_0^+ e^{-j\beta z} - V_0^- e^{j\beta z}\right)$$

Step 2: Apply the short

At $z = 0$, $V(0) = 0$:

$$V_0^+ + V_0^- = 0 \;\Rightarrow\; V_0^- = -V_0^+$$

So $\Gamma_L = -1$.

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Step 3: Voltage

$$V(z) = V_0^+ \left(e^{-j\beta z} - e^{j\beta z}\right)$$

Using $e^{-jx} - e^{jx} = -2j\sin x$:

$$\boxed{V(z) = -2jV_0^+ \sin \beta z}$$

Step 4: Current

$$I(z) = \frac{V_0^+}{Z_0}\left(e^{-j\beta z} + e^{j\beta z}\right)$$

Using $e^{-jx} + e^{jx} = 2\cos x$:

$$\boxed{I(z) = \frac{2V_0^+}{Z_0}\cos \beta z}$$

Step 5: Impedance a distance l from the short

Put $z = -l$. Since $\sin(-\beta l) = -\sin \beta l$ and $\cos(-\beta l) = \cos \beta l$:

$$Z_{in} = \frac{V(-l)}{I(-l)} = \frac{2jV_0^+ \sin \beta l}{(2V_0^+/Z_0)\cos \beta l}$$

$$\boxed{Z_{in} = jZ_0 \tan \beta l}$$

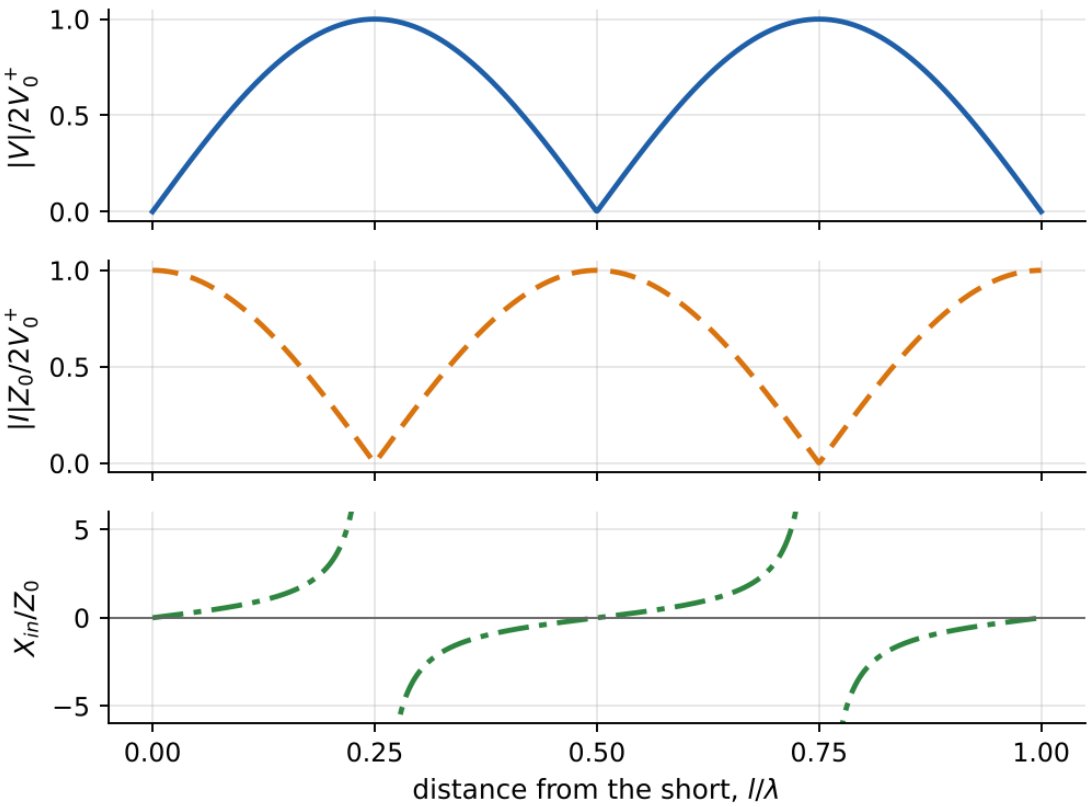


Figure 17: Voltage, current and reactance along a short-circuited line

Reading the figure. Top: the voltage is zero at the short and at every $\lambda/2$ from it. Middle: the current is largest at the short and zero at $\lambda/4$. Bottom: the reactance starts at zero, is inductive up to $\lambda/4$, goes through infinity there (an open circuit), is capacitive up to $\lambda/2$, and then repeats.

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Steps to remember

General solution → Apply the short → Voltage → Current → Impedance a distance l from the short.

4.(d) How is quarter wavelength transmission line used for impedance matching? Marks: 6

1. Theory

1. Why matching matters (Core Idea)

When a line of impedance Z_0 feeds a load $R_L eZ_0$, part of the wave is reflected: less power reaches the load and a standing wave appears on the line. We want something between the line and the load that makes the load "look like" Z_0 from the line, without wasting power. A short piece of a *different* line can do this.

2. A line changes the impedance it shows

A load Z_L seen through a lossless line of impedance Z_1 and length l looks like (derived in 3(b))

$$Z_{in} = Z_1 \frac{Z_L + jZ_1 \tan \beta l}{Z_1 + jZ_L \tan \beta l}$$

So by choosing the length and the impedance of the piece, we choose what the line sees.

3. The special length lambda/4

For $l = \lambda/4$, $\beta l = \pi/2$ and $\tan \beta l$ becomes infinite. Dividing the top and bottom by $\tan \beta l$ leaves only the j terms, and the formula becomes $Z_{in} = Z_1^2/Z_L$: the quarter-wave line turns an impedance "upside down" (a small Z_L looks large, a large one small). Another way to see it: the wave reflected at the far end of the section travels $\lambda/4$ there and $\lambda/4$ back, half a wavelength in all, so it returns turned by 180° and cancels the reflection from the near end, when Z_1 has the right value.

2. Answer

A section of line of characteristic impedance Z_1 and length $l = \lambda/4$ is placed between the main line (Z_0) and a resistive load R_L. Its input impedance is

$$Z_{in} = Z_1 \frac{R_L + jZ_1 \tan \beta l}{Z_1 + jR_L \tan \beta l}$$

With $\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$, $\tan \beta l \to \infty$, so

$$Z_{in} = \frac{Z_1^2}{R_L}$$

For a match we need $Z_{in} = Z_0$:

$$\boxed{Z_1 = \sqrt{Z_0 R_L}}$$

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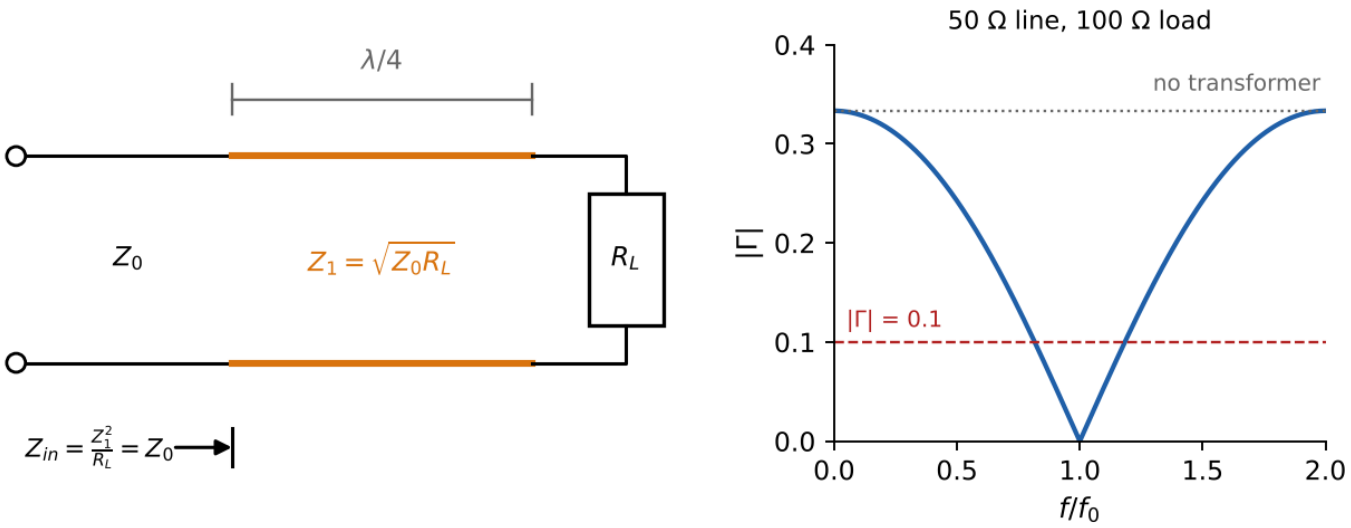


Figure 18: Quarter-wave transformer and its bandwidth

Reading the figure. Left: the thick orange $\lambda/4$ section of impedance $\sqrt{Z_0 R_L}$ between the line and R_L. Right: the reflection seen from the line for a 50 Ω line and a 100 Ω load (solid blue): zero at the design frequency f_0, rising on both sides.

Key Points: it works exactly only at one frequency (narrow band); the load must be real. A complex load is first moved along the line to a voltage maximum or minimum, where the impedance is real ($Z_0 S$ or Z_0/S), and the transformer is put there. Example: a 100 Ω load on a 50 Ω line needs $Z_1 = \sqrt{50 \times 100} = 70.7\ \Omega$.

3. Final Concise Answer

(a) R with C_a in parallel and lead inductance L in series; $|Z|$ stays at R, falls after $f_c = 1/(2\pi RC_a)$, has a minimum at the series resonance, then rises with the lead reactance.

(b) A half-wavelength metal patch over a ground plane, radiating from its edges. Microstrip lines are used in MICs, matching networks, filters, couplers and array feeds.

(c) $V = -2jV_0^+ \sin\beta z$, $I = (2V_0^+/Z_0)\cos\beta z$, $Z_{in} = jZ_0 \tan\beta l$.

(d) A $\lambda/4$ section of $Z_1 = \sqrt{Z_0 R_L}$ matches a real load R_L to the line Z_0.

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5. (a) For high pass filter determine ABCD parameters of a two port system. Marks: 6
   (b) What are stability circles? Explain how it is used. Marks: 5
   (c) Explain the working principle of negative resistance oscillator and feedback oscillator. Marks: 7
   (d) A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6\ \Omega$ and short circuit impedance $j103\ \Omega$. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input. Marks: 7

5.(a) For high pass filter determine ABCD parameters of a two port system. Marks: 6

1. Theory

1. What is a two-port? (Core Idea)

A two-port is a box with an input pair of terminals and an output pair. We do not need to know the inside. A few numbers relate the voltages and currents at the two ports.

2. What are ABCD parameters?

They give the input in terms of the output:

$$V_1 = AV_2 + BI_2, \qquad I_1 = CV_2 + DI_2$$

where I_2 flows out of port 2. Their advantage: when two-ports are in a chain, the output of one is the input of the next, so the ABCD matrices simply multiply.

3. What does a high-pass filter do?

It passes high frequencies and stops low ones. A capacitor in series blocks low frequencies, since $1/\omega C$ is large there. An inductor to ground shorts low frequencies away, since ωL is small.

2. Answer

Assumption: the paper gives no circuit, so the basic LC high-pass section is used: series C, then shunt L. Let $Z = 1/(j\omega C)$ and $Y = 1/(j\omega L)$.

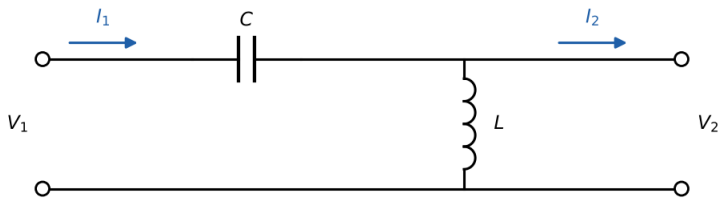


Figure 19: LC high-pass section

Step 1: Series element

Across a series Z: $V_1 = V_2 + ZI_2$ and $I_1 = I_2$. So

$$\begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

Step 2: Shunt element

Across a shunt Y: $V_1 = V_2$ and $I_1 = YV_2 + I_2$. So

$$\begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$$

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Step 3: Cascade

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix} = \begin{bmatrix} 1+ZY & Z \\ Y & 1 \end{bmatrix}$$

Step 4: Put in the elements

$$ZY = \frac{1}{j\omega C} \cdot \frac{1}{j\omega L} = -\frac{1}{\omega^2 LC}$$

$$\boxed{A = 1 - \frac{1}{\omega^2 LC}, \quad B = \frac{1}{j\omega C}, \quad C = \frac{1}{j\omega L}, \quad D = 1}$$

Here C on the left is the ABCD parameter and C on the right is the capacitor.

Check: a reciprocal network has $AD - BC = 1$. Here $(1 + ZY)(1) - ZY = 1$ ✔.

Behaviour: as $\omega \to \infty$, $A \to 1$ and $B, C \to 0$: the section passes the signal unchanged. As $\omega \to 0$, $|A|$ and $|B|$ grow without limit: nothing passes.

Steps to remember

Series element → Shunt element → Cascade → Put in the elements.

3. Notes and Remarks

- For the RC high-pass (series C, shunt R), put $Y = 1/R$: $A = 1 + 1/(j\omega RC)$, $B = 1/(j\omega C)$, $C = 1/R$, $D = 1$.

5.(b) What are stability circles? Explain how it is used. Marks: 5

1. Theory

1. What does stability mean in an amplifier? (Core Idea)

An amplifier must amplify only the signal given to it. If some of its output gets back to its input in the right phase, it can start oscillating by itself. In terms of waves, this happens if a port reflects more than it receives: $|\Gamma_{in}| > 1$ or $|\Gamma_{out}| > 1$.

2. Why do the source and load matter?

The transistor has $S_{12} \neq 0$, so a little of the output gets back to the input. Its input reflection depends on the load:

$$\Gamma_{in} = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$$

So some loads are safe and some are not. The same holds for Γ_{out} and the source.

3. Why a circle?

The formula above is a fraction with Γ_L on the top and bottom. Such a function maps circles to circles. So the boundary $|\Gamma_{in}| = 1$ is a circle on the Γ_L plane.

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2. Answer

Stability circle: a circle on the Smith chart (Γ_L or Γ_S plane) that separates the loads or sources which keep the transistor stable from those which can make it oscillate. It is the set of points where $|\Gamma_{in}| = 1$ or $|\Gamma_{out}| = 1$.

Output stability circle (for Γ_L), with $\Delta = S_{11}S_{22} - S_{12}S_{21}$:

$$C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{|S_{22}|^2 - |\Delta|^2}, \qquad R_L = \left| \frac{S_{12}S_{21}}{|S_{22}|^2 - |\Delta|^2} \right|$$

Input stability circle (for Γ_S):

$$C_S = \frac{(S_{11} - \Delta S_{22}^*)^*}{|S_{11}|^2 - |\Delta|^2}, \qquad R_S = \left| \frac{S_{12}S_{21}}{|S_{11}|^2 - |\Delta|^2} \right|$$

How it is used:

1. Compute the centre and radius from the transistor's S-parameters at the working frequency.
2. Draw the circle on the Smith chart.
3. Find which side is stable. Test the chart's centre, $\Gamma_L = 0$, where $\Gamma_{in} = S_{11}$. If $|S_{11}| < 1$, the centre is stable, and so is the whole side containing it.
4. Choose Γ_L and Γ_S in the stable region only.

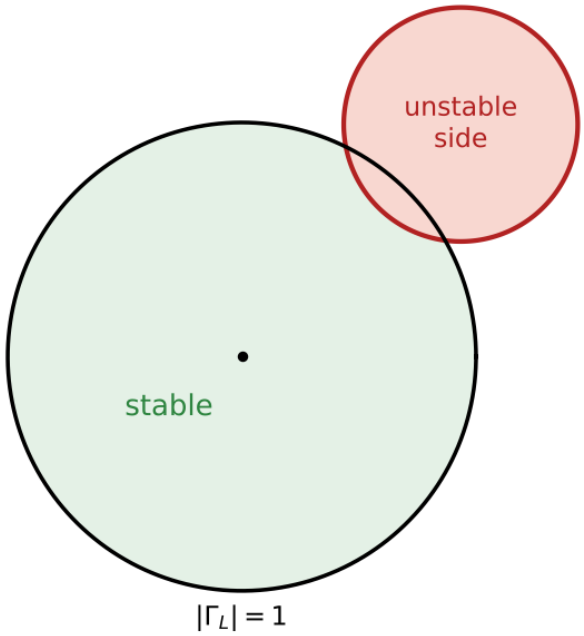


Figure 20: An output stability circle on the Γ_L plane

Reading the figure. The black circle is the Smith chart's edge. The red circle has $C_L = 1.36\angle 46.7°$ and $R_L = 0.50$, for an illustrative transistor ($S_{11} = 0.894\angle -60.6°$, $S_{12} = 0.02\angle 62.4°$, $S_{21} = 3.122\angle 123.6°$, $S_{22} = 0.781\angle -27.6°$). Since $|S_{11}| < 1$, the side containing the chart's centre is stable. Loads inside the red circle must be avoided.

Unconditional stability: if the circles lie fully outside the chart, every passive load is stable. The Rollett test gives this directly: $K > 1$ and $|\Delta| < 1$. This example has $K = 0.61$, so it is only conditionally stable.

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5.(c) Explain the working principle of negative resistance oscillator and feedback oscillator. Marks: 7

1. Theory

1. What is an oscillator? (Core Idea)

An oscillator makes a sine wave from DC power, with no input signal. A real resonant circuit loses energy each cycle, so its ringing dies. To keep it going, something must put the lost energy back, once each cycle, in step.

2. How does it start?

Random noise is always present. The circuit picks out the frequency at which the loop adds in phase. That small signal grows. The gain then drops as the device saturates, and the amplitude settles.

2. Answer

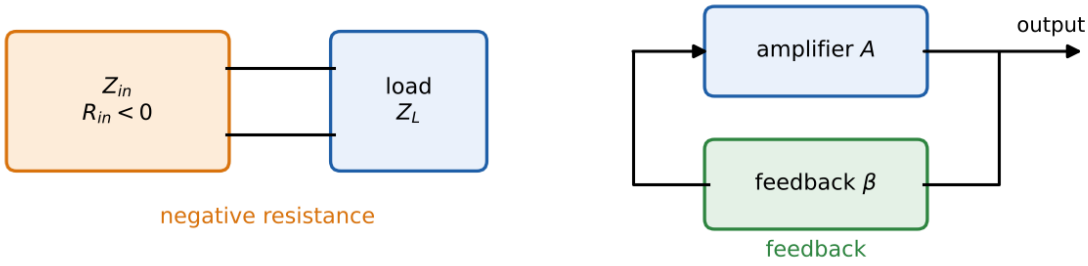


Figure 21: The two kinds of oscillator

Reading the figure. Left: an active one-port with negative input resistance, connected to a load. Right: an amplifier A with a feedback network β returning part of its output to its input.

Negative resistance oscillator

A positive resistance takes power. A **negative resistance** gives power: the voltage falls as the current rises, as in a tunnel or Gunn diode, or a transistor with suitable feedback. Connect a one-port of impedance $Z_{in} = R_{in} + jX_{in}$, with $R_{in} < 0$, to a passive load $Z_L = R_L + jX_L$.
The loop has current I and no source, so KVL gives

$$(Z_{in} + Z_L)I = 0$$

For a non-zero current, $Z_{in} + Z_L = 0$. Taking the real and imaginary parts:

$$\boxed{R_{in} + R_L = 0, \qquad X_{in} + X_L = 0}$$

The negative resistance cancels the load's loss, and the reactances resonate at the oscillation frequency. For oscillation to start, $|R_{in}|$ must exceed R_L at first, usually with $R_L = |R_{in}|/3$. As the amplitude grows, $|R_{in}|$ falls until the two are equal.

Feedback oscillator

An amplifier of gain A has part β of its output fed back, in phase, to its input. With an input V_i, the output is $V_o = A(V_i + \beta V_o)$, so

$$\frac{V_o}{V_i} = \frac{A}{1 - A\beta}$$

If $A\beta = 1$, the gain is infinite, which means an output with no input. This is the **Barkhausen condition**:

$$\boxed{|A\beta| = 1, \qquad \angle A\beta = 0° \text{ or } 360°}$$

To start, $|A\beta|$ must be slightly above 1. Saturation then brings it to 1. The frequency is set by an LC

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or crystal network in β, as in the Colpitts and Hartley oscillators.

5.(d) A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6\ \Omega$ and short circuit impedance $j103\ \Omega$. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input. Marks: 7

1. Theory

1. Why measure open and short? (Core Idea)

A line's Z_0 and β cannot be read from a simple meter. But the input impedances with the far end open and shorted can. Two measurements give two unknowns.

2. Where do the formulas come from?

From the general input impedance of a lossless line,

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$$

Short, $Z_L = 0$: $Z_{sc} = Z_0 \dfrac{jZ_0 \tan \beta l}{Z_0} = jZ_0 \tan \beta l.$

Open, $Z_L \to \infty$: divide the top and bottom by Z_L. Only the terms with Z_L survive: $Z_{oc} = Z_0 \dfrac{1}{j \tan \beta l} = -jZ_0 \cot \beta l.$

2. Answer

Given: $l = 1.5$ m $< \lambda/4$; $Z_{oc} = -j54.6\ \Omega$; $Z_{sc} = j103\ \Omega$; lossless.

Step 1: Characteristic impedance

Multiply the two:

$$Z_{oc} Z_{sc} = (-jZ_0 \cot \beta l)(jZ_0 \tan \beta l) = Z_0^2$$

$$Z_0 = \sqrt{(-j54.6)(j103)} = \sqrt{5623.8}$$

$$\boxed{Z_0 = 75.0\ \Omega}$$

Step 2: Propagation constant

Divide the two:

$$\frac{Z_{sc}}{Z_{oc}} = \frac{jZ_0 \tan \beta l}{-jZ_0 \cot \beta l} = -\tan^2 \beta l$$

$$\tan^2 \beta l = \frac{103}{54.6} = 1.886 \ \Rightarrow\ \tan \beta l = 1.373$$

The sign is positive because the line is shorter than $\lambda/4$, so $\beta l < 90°$:

$$\beta l = 53.9° = 0.941 \text{ rad}, \qquad \beta = \frac{0.941}{1.5} = 0.628 \text{ rad/m}$$

$$\boxed{\gamma = j\beta = j0.628 \text{ m}^{-1}}$$

So $\lambda = 2\pi/\beta = 10.0$ m, and $\lambda/4 = 2.5$ m > 1.5 m ✔.

Step 3: Input impedance with the length doubled

Now $l = 3$ m and $2\beta l = 107.9°$, with $\tan 107.9° = -3.10$. The paper does not say which end is meant, so both are given.

$$Z'_{sc} = jZ_0 \tan(2\beta l) = j75(-3.10)$$

$$\boxed{Z'_{sc} = -j232\ \Omega}$$

$$Z'_{oc} = -jZ_0 \cot(2\beta l) = -j75\left(\frac{1}{-3.10}\right)$$

$$\boxed{Z'_{oc} = +j24.2\ \Omega}$$

Step 4: Length that looks like an open circuit

A **short-circuited** line looks open when $\tan \beta l \to \infty$, so $\beta l = \pi/2$:

$$l = \frac{\lambda}{4} = \frac{10.0}{4}$$

$$\boxed{l = 2.5 \text{ m} \quad (\text{and every } \lambda/2 = 5 \text{ m further})}$$

An **open-circuited** line looks open at $l = \lambda/2 = 5$ m, and every 5 m further.

Final Answer: $Z_0 = 75\ \Omega$; $\gamma = j0.628\ \text{m}^{-1}$; doubled length: $-j232\ \Omega$ (short end) or $+j24.2\ \Omega$ (open end); $l = 2.5$ m with a short end.

3. Final Concise Answer

(a) Ideal series C, shunt L: $A = 1 - 1/(\omega^2 LC)$, $B = 1/(j\omega C)$, $C = 1/(j\omega L)$, $D = 1$.

(b) Circles on the Γ_L or Γ_S plane where $|\Gamma_{in}| = 1$ or $|\Gamma_{out}| = 1$; choose loads and sources only on the stable side.

(c) Negative resistance: $R_{in} + R_L = 0$, $X_{in} + X_L = 0$. Feedback: $|A\beta| = 1$ with zero loop phase.

(d) $Z_0 = 75\ \Omega$, $\gamma = j0.628\ \text{m}^{-1}$; doubled: $-j232\ \Omega$ (short) or $+j24.2\ \Omega$ (open); $l = 2.5$ m.

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6. (a) Write short notes on the following topics.

   (i) Poincare Sphere
   (ii) Yagi-Uda antenna. Marks: 6

   (b) For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio. Marks: 6

   (c) Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them. Marks: 6

   (d) For a pyramidal or rectangular horn show that directivity D is $D \approx 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right)$. Marks: 7

6.(a) Write short notes on the following topics. (i) Poincare Sphere (ii) Yagi-Uda antenna. Marks: 6

1. Theory

1. What is polarization? (Core Idea)

The electric field of a wave points in some direction at each moment. The path the tip of that vector traces, seen along the wave, is its **polarization**: a line (linear), a circle (circular), or generally an ellipse (elliptical).

2. How can one antenna have many rods but one feed?

A rod near a driven rod picks up its wave and radiates again. This is a **parasitic** element. The phase of its current depends on whether it is slightly longer or shorter than resonance, so the rods can be arranged to add forwards and cancel backwards.

2. Answer

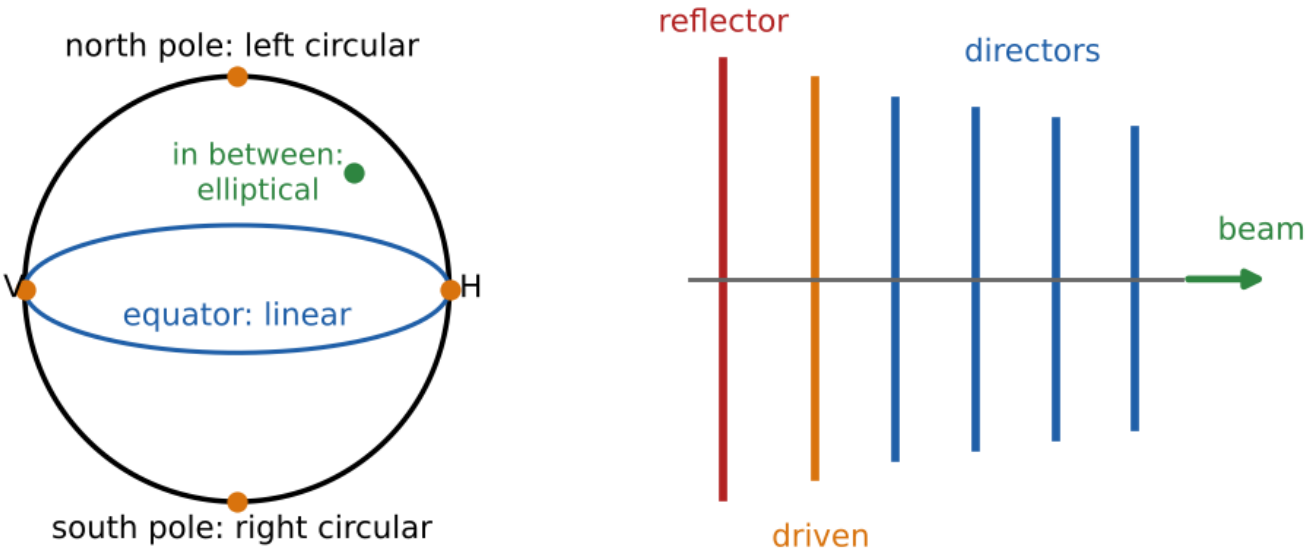


Figure 22: Left: the Poincaré sphere. Right: a Yagi-Uda antenna

Reading the figure. Left: each polarization is one point on the sphere; here the circular states are at the poles, linear ones on the equator (H and V opposite each other), and elliptical ones elsewhere. Right: one red reflector, one orange driven element and several blue directors, with the beam going towards the directors.

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i. Poincaré sphere

A sphere on which every polarization state is one point. An ellipse has an orientation angle τ and an ellipticity angle ε. The point has longitude 2τ and latitude 2ε.

- **Equator:** linear polarization.
- **Poles:** left and right circular (which is at which pole depends on the convention).
- **Elsewhere:** elliptical, with left-handed and right-handed in opposite hemispheres.
- **Opposite points:** orthogonal polarizations, such as horizontal and vertical.

Use: the polarization match between a wave and an antenna depends on the angle 2γ between their points on the sphere:

$$F = \cos^2 \gamma$$

Same point: $F = 1$, all the power received. Opposite points: $F = 0$, none.

ii. Yagi-Uda antenna

An end-fire array of parallel dipoles, with only one of them fed.

- **Driven element:** a resonant dipole, about $\lambda/2$, connected to the line.
- **Reflector:** about 5 % longer, behind the driven element. It is inductive, and it sends the wave forward.
- **Directors:** several, shorter than the driven element, in front. They are capacitive, and they lead the wave on along the array.

Properties: gain about 7 to 15 dBi, growing with the number of directors; simple and cheap; narrow band. Its mutual coupling lowers the feed impedance, so a folded dipole is often used as the driven element.
Use: TV reception, amateur radio and point-to-point VHF and UHF links.

6.(b) For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio. Marks: 6

1. Theory

1. What does a paraboloid do? (Core Idea)

A **paraboloid of revolution** is a parabola turned round its axis, like a satellite dish. It has one special property: every ray from its focus reflects into a direction parallel to the axis, and every such path to the mouth has the same length. So a spherical wave from the focus leaves as a plane wave.

2. Why is the edge weaker than the centre?

The rim is further from the focus than the vertex. The wave has spread more on the way there. Also, the ray tubes hit the dish at a slant. So the aperture is not lit evenly: the centre is bright and the edge is dim. This is called **taper**.

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2. Answer

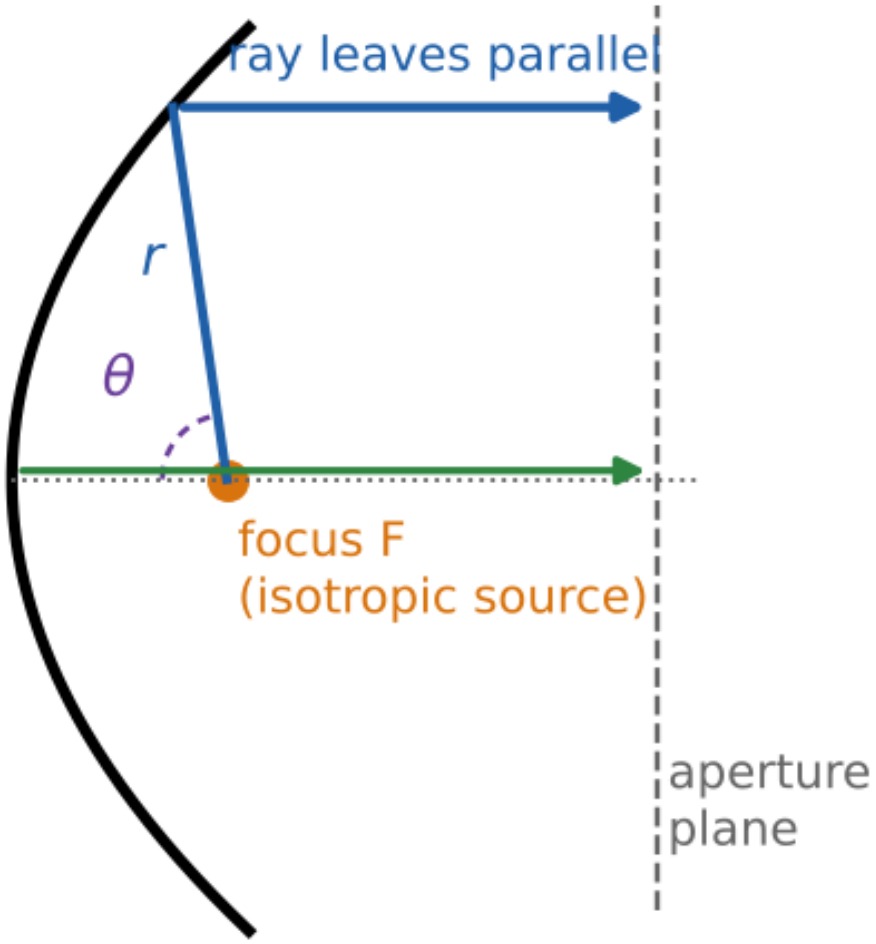


Figure 23: A paraboloid with an isotropic source at the focus

Reading the figure. The source at the focus F is a distance f from the vertex. A ray at angle θ from the axis, measured towards the vertex, travels a distance r to the dish and leaves parallel to the axis.
Set-up. Isotropic source of intensity U at the focus. Focal length f.

Step 1: Shape of the dish

The parabola's definition gives the distance from the focus:

$$r = \frac{2f}{1+\cos\theta}$$

At the vertex, $\theta = 0$ and $r = f$. Using $1+\cos\theta = 2\cos^2(\theta/2)$, $r = f\sec^2(\theta/2)$.

Step 2: Radius on the aperture

A ray leaves at the distance from the axis

$$\rho = r\sin\theta = \frac{2f\sin\theta}{1+\cos\theta} = 2f\tan\frac{\theta}{2}$$

using $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$. Differentiating:

$$\frac{d\rho}{d\theta} = f\sec^2\frac{\theta}{2} = r$$

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Step 3: Power balance

The power in the cone between θ and $\theta + d\theta$ is $U \cdot 2\pi \sin\theta \, d\theta$. After reflection it passes through a ring of the aperture of area $2\pi\rho \, d\rho$, with density $S(\theta)$. Rays do not cross, so the two are equal:

$$S(\theta)\, 2\pi\rho \, d\rho = U \, 2\pi \sin\theta \, d\theta$$

With $\rho = r\sin\theta$ and $d\rho = r \, d\theta$, the left side is $S(\theta)\, 2\pi r^2 \sin\theta \, d\theta$:

$$S(\theta) = \frac{U}{r^2}$$

Step 4: Ratio to the centre

At the centre, $r = f$ and $S(0) = U/f^2$. So

$$\frac{S(\theta)}{S(0)} = \frac{f^2}{r^2}$$

$$\boxed{\frac{S(\theta)}{S(0)} = \left(\frac{1+\cos\theta}{2}\right)^2 = \cos^4\frac{\theta}{2}}$$

Step 5: Field intensity ratio

The power density goes as the square of the field, so

$$\boxed{\frac{E(\theta)}{E(0)} = \frac{1+\cos\theta}{2} = \cos^2\frac{\theta}{2}}$$

Example

at $\theta = 60°$, $S/S(0) = 0.5625$ and $E/E(0) = 0.75$, which is 2.5 dB down.

Steps to remember

Shape of the dish → Radius on the aperture → Power balance → Ratio to the centre → Field intensity ratio.

6.(c) Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them. Marks: 6

1. Theory

1. What does an aperture collect? (Core Idea)

A receiving antenna behaves like a net: it collects the power falling on its effective area A_e. A transmitting antenna of area A also focuses its beam, and a larger area in wavelengths gives a narrower beam.

2. How does directivity connect to area?

The link between them is

$$D = \frac{4\pi A_e}{\lambda^2}$$

So a shorter wavelength makes the same area more directive.

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2. Answer

Set-up. Transmitting and receiving antennas of effective aperture A_{et} and A_{er}, a distance r apart, in the far field, matched and lossless.

Step 1: Density from an isotropic source

$$S_{iso} = \frac{P_t}{4\pi r^2}$$

Step 2: Density with the directive antenna

$$S = D_t S_{iso} = \frac{P_t D_t}{4\pi r^2}$$

Step 3: Put in the aperture

With $D_t = 4\pi A_{et}/\lambda^2$, the 4π cancels:

$$S = \frac{P_t}{4\pi r^2} \cdot \frac{4\pi A_{et}}{\lambda^2} = \frac{P_t A_{et}}{r^2 \lambda^2}$$

Step 4: Received power

$$P_r = S A_{er} = \frac{P_t A_{et} A_{er}}{r^2 \lambda^2}$$

Step 5: Ratio

$$\frac{P_r}{P_t} = \frac{A_{et} A_{er}}{r^2 \lambda^2}$$

For the same aperture A on both:

$$\boxed{\frac{P_r}{P_t} = \frac{A^2}{r^2 \lambda^2} \propto \frac{1}{\lambda^2 r^2}}$$

So the ratio is inversely proportional to the square of the wavelength and to the square of the distance, as required.

Steps to remember

Density from an isotropic source → Density with the directive antenna → Put in the aperture → Received power → Ratio.

Interpretation: with fixed apertures a higher frequency gives a stronger link, because the beams are narrower. With fixed gains the opposite holds, as in $P_r/P_t = G_t G_r (\lambda/4\pi r)^2$, because the areas shrink.

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6.(d) For a pyramidal or rectangular horn show that directivity D is $D \approx 10\log\left(\frac{7.5A_p}{\lambda^2}\right)$.

Marks: 7

1. Theory

1. What is a horn? (Core Idea)

A horn is a waveguide whose end opens out gradually. It matches the guide to free space and gives a larger aperture, so a narrower beam. A **pyramidal** horn flares in both the E and H planes, and its mouth is a rectangle of area A_p.

2. Why is the effective area less than the physical area?

The wave does not light the mouth evenly, since the waveguide's field falls to zero at the side walls. It also reaches the edges a little later than the centre, because the path is longer, so the phase is not uniform. Both lower the effective area A_e below the physical area A_p.

$$\epsilon_{ap} = \frac{A_e}{A_p}$$

is the **aperture efficiency**.

2. Answer

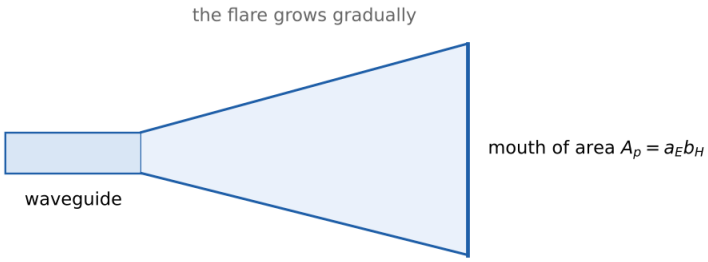


Figure 24: A pyramidal horn

Step 1: Directivity and effective aperture

For any antenna,

$$D = \frac{4\pi A_e}{\lambda^2}$$

Step 2: Effective and physical aperture

$$A_e = \epsilon_{ap} A_p$$

so

$$D = \frac{4\pi \epsilon_{ap} A_p}{\lambda^2}$$

Step 3: Aperture efficiency of a horn

A well-designed horn has $\epsilon_{ap} \approx 0.6$. The uneven amplitude and the phase error across the mouth account for the loss. So

$$4\pi\epsilon_{ap} = 4\pi \times 0.6 = 7.54 \approx 7.5$$

$$D \approx \frac{7.5A_p}{\lambda^2}$$

Step 4: In decibels

$$\boxed{D \approx 10\log_{10}\left(\frac{7.5A_p}{\lambda^2}\right) \text{ dBi}}$$

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as required.

Example

a horn with $A_p = 10\lambda^2$ has $D = 75$, which is 18.8 dBi.

Steps to remember

Directivity and effective aperture → Effective and physical aperture → Aperture efficiency of a horn → In decibels.

3. Final Concise Answer

(a) Poincaré sphere: one point for each polarization, with linear on the equator and circular at the poles. Yagi-Uda: an end-fire array with one driven dipole, one reflector and several directors.

(b) $S(\theta)/S(0) = [(1+\cos\theta)/2]^2$ and $E(\theta)/E(0) = (1+\cos\theta)/2$.

(c) $P_r/P_t = A^2/(r^2\lambda^2)$.

(d) $D = 4\pi\epsilon_{ap}A_p/\lambda^2$ with $\epsilon_{ap} \approx 0.6$ gives $D \approx 7.5A_p/\lambda^2$, or $10\log_{10}(7.5A_p/\lambda^2)$ dBi.

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FORMULA SHEET

EEE475

RF and Microwave Engineering

Formula Sheet

Session 2018–19 paper

Every formula behind the 4th Year 1st Semester Final Examination, 2022

TOPICS

1 Transmission lines — 8 formulas

2 Matching — 3 formulas

3 RF components and two-ports — 4 formulas

4 Stability and oscillators — 5 formulas

5 Antennas — 11 formulas

6 Arrays — 5 formulas

7 Links, radar and remote sensing — 6 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

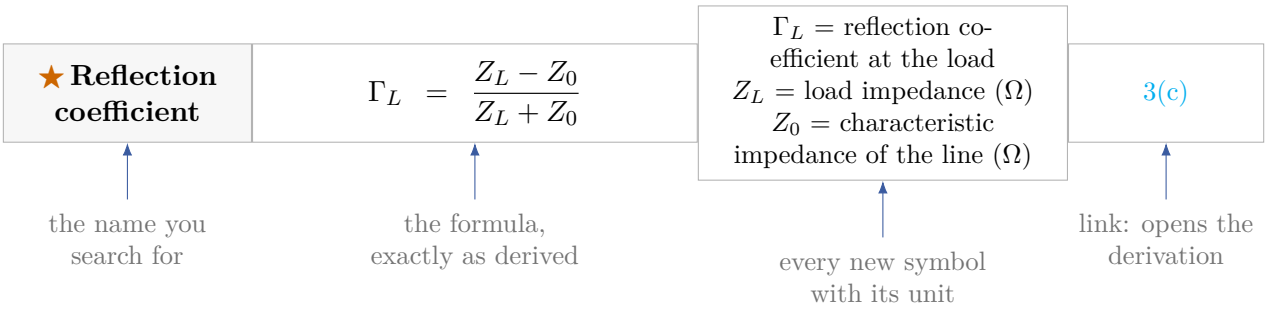
Prepared by Nur Mohammad

October 2026

BACK OF THE COVER

Before you use the sheet

How to read a row



In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

Contents

1 Transmission lines **4**

2 Matching **5**

3 RF components and two-ports **5**

4 Stability and oscillators **6**

5 Antennas **6**

6 Arrays **7**

7 Links, radar and remote sensing **8**

8 Key formulas **9**

9 Symbol index **10**

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Conventions (Pozar for lines, amplifiers and radar; Ludwig and Bretchko for RF components; Kraus for antennas): phasors with $e^{j\omega t}$; the load is at $z = 0$; $k = \beta = 2\pi/\lambda$.

Columns: Title · Equation · Notation · Ref. (reference).

1. Transmission lines:

Title	Equation	Notation	Ref.
★ **Reflection coefficient**	$\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$	Γ_L = reflection coefficient at the load Z_L = load impedance (Ω) Z_0 = characteristic impedance of the line (Ω)	3(c)
★ **Standing wave ratio**	$S = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	S = standing wave ratio, V_{max}/V_{min} Γ = reflection coefficient	3(c)
Voltage maximum	$d_{max} = \dfrac{\theta\lambda}{4\pi}$	d_{max} = distance of the first voltage maximum from the load (m) θ = angle of Γ_L (rad) λ = wavelength on the line (m)	3(c)
★ **Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	Z_{in} = input impedance (Ω) $j = \sqrt{-1}$ β = phase constant (rad/m) l = line length (m)	3(b), 3(c)
Short-circuited line	$V = -2jV_0^+ \sin\beta z,\ I = \dfrac{2V_0^+}{Z_0} \cos\beta z$	V, I = voltage and current on the line (V, A) V_0^+ = incident amplitude (V) z = position, the load at $z = 0$ (m)	4(c)
★ **Short and open input**	$Z_{sc} = jZ_0 \tan\beta l,\ Z_{oc} = -jZ_0 \cot\beta l$	Z_{sc}, Z_{oc} = input impedances with the end shorted and open (Ω) reactive for any length	4(c), 5(d)
Z_0 from two measurements	$Z_0 = \sqrt{Z_{oc} Z_{sc}}$	—	5(d)
βl from two measurements	$\tan^2 \beta l = -Z_{sc}/Z_{oc}$	—	5(d)

Table 1: Transmission lines

2. Matching:

Title	Equation	Notation	Ref.
★ **Quarter-wave transformer**	$Z_1 = \sqrt{Z_0 R_L}$	Z_1 = section impedance (Ω) Z_0 = line impedance (Ω) R_L = real load (Ω)	4(d)
Stub position	$t = \tan\beta d = \dfrac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]/Z_0}}{R_L - Z_0}$	$t = \tan\beta d$ β = phase constant (rad/m) d = distance from the load (m) X_L = load reactance (Ω)	3(d)
Short-circuited stub	$\dfrac{l}{\lambda} = \dfrac{1}{2\pi}\tan^{-1}\dfrac{Y_0}{B}$	l = stub length (m) λ = wavelength (m) $Y_0 = 1/Z_0$, line admittance (S) B = susceptance to cancel (S)	3(d)

Table 2: Matching

3. RF components and two-ports:

Title	Equation	Notation	Ref.
High-frequency resistor	$Z = j\omega(2L) + \dfrac{1}{1/R + j\omega C_a}$	Z = impedance of the resistor (Ω) $j = \sqrt{-1}$ ω = angular frequency (rad/s) L = each lead's inductance (H) R = resistance (Ω) C_a = end capacitance (F)	4(a)
Resistor corners	$f_c = \dfrac{1}{2\pi R C_a},\ f_0 = \dfrac{1}{2\pi\sqrt{2LC_a}}$	f_c = corner where C_a starts to short R (Hz) f_0 = series resonance of the leads with C_a (Hz)	4(a)
ABCD definition	$V_1 = AV_2 + BI_2,\ I_1 = CV_2 + DI_2$	V_1, I_1 = voltage and current at port 1 V_2, I_2 = voltage and current at port 2 A, B, C, D = ABCD parameters I_2 flows out of port 2	5(a)
★ **High-pass ABCD**	$A = 1 - \dfrac{1}{\omega^2 LC},\ B = \dfrac{1}{j\omega C},\ C = \dfrac{1}{j\omega L},\ D = 1$	C = series capacitance (F) L = shunt inductance (H) series C, shunt L	5(a)

Table 3: RF components and two-ports

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4. Stability and oscillators:

Title	Equation	Notation	Ref.
Input reflection	$\Gamma_{in} = S_{11} + \dfrac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$	Γ_{in} = reflection at the input S_{11}, S_{12}, S_{21}, S_{22} = S-parameters of the transistor Γ_L = load reflection coefficient	5(b)
Output stability circle	$C_L = \dfrac{(S_{22} - \Delta S_{11}^*)^*}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2},\ R_L = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	C_L, R_L = centre and radius of the output stability circle $\Delta = S_{11}S_{22} - S_{12}S_{21}$ S_{11}^*, S_{22}^* = complex conjugates of S_{11}, S_{22}	5(b)
Input stability circle	$C_S = \dfrac{(S_{11} - \Delta S_{22}^*)^*}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2},\ R_S = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	C_S, R_S = centre and radius of the input stability circle	5(b)
Negative-resistance oscillator	$R_{in} + R_L = 0,\ X_{in} + X_L = 0$	R_{in}, X_{in} = input resistance and reactance of the device (Ω) R_L, X_L = load resistance and reactance (Ω) $R_{in} < 0$	5(c)
★ **Barkhausen condition**	$\lvert A\beta\rvert = 1,\ \angle A\beta = 0°$	A = gain β = feedback fraction	5(c)

Table 4: Stability and oscillators

5. Antennas:

Title	Equation	Notation	Ref.
★ **Directivity from aperture**	$D = \dfrac{4\pi A_e}{\lambda^2}$	D = directivity A_e = effective aperture (m^2) λ = wavelength (m)	1(a), 6(c), 6(d)
Power and beam area	$P_{rad} = \iint U\, d\Omega,\ \Omega_A = \dfrac{P_{rad}}{U_m}$	P_{rad} = radiated power (W) U = radiation intensity (W/sr) Ω = solid angle (sr) Ω_A = beam area (sr) U_m = largest radiation intensity (W/sr)	1(b)
★ **Directivity**	$D = \dfrac{4\pi U_m}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$	—	1(b)
Field regions	$r_1 = 0.62\sqrt{D^3/\lambda},\ r_2 = 2D^2/\lambda$	r_1, r_2 = inner and outer limits of the radiating near field (m) D = largest dimension of the antenna (m)	1(c)
Magnetic dipole E	$E_\phi = \dfrac{\eta k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	E_ϕ = electric field (V/m) η = wave impedance of free space (377 Ω) $k = 2\pi/\lambda$ (rad/m) I = loop current (A) S = loop area (m^2) r = distance (m) θ = angle from the loop axis $j = \sqrt{-1}$	1(d)

Title	Equation	Notation	Ref.
Magnetic dipole H_r	$H_r = \frac{jkIS}{2\pi r^2}\cos\theta\left[1+\frac{1}{jkr}\right]e^{-jkr}$	H_r = radial magnetic field (A/m)	1(d)
Magnetic dipole H_θ	$H_\theta = -\frac{k^2IS}{4\pi r}\sin\theta\left[1+\frac{1}{jkr}-\frac{1}{(kr)^2}\right]e^{-jkr}$	H_θ = θ component of the magnetic field (A/m)	1(d)
Paraboloid shape	$r = \frac{2f}{1+\cos\theta}$	r = distance from the focus to the dish (m) f = focal length (m) θ = angle from the axis	6(b)
★ **Paraboloid taper**	$\frac{S(\theta)}{S(0)} = \left(\frac{1+\cos\theta}{2}\right)^2$, $\frac{E(\theta)}{E(0)} = \frac{1+\cos\theta}{2}$	$S(\theta)$ = power density at the aperture (W/m^2) $E(\theta)$ = field at the aperture (V/m)	6(b)
★ **Horn directivity**	$D \approx \frac{7.5A_p}{\lambda^2}$	A_p = mouth area (m^2) $\epsilon_{ap} \approx 0.6$	6(d)
Polarization match	$F = \cos^2\gamma$	F = polarization match factor γ = half the angle between the two polarizations on the Poincaré sphere	6(a)

Table 5: Antennas

6. Arrays:

Title	Equation	Notation	Ref.
Phase difference	$\psi = \beta d\cos\theta + \delta$	ψ = phase difference between neighbouring elements (rad) $\beta = 2\pi/\lambda$ (rad/m) d = spacing (m) θ = angle from the array axis δ = feed phase (rad)	2(d)
★ **Array factor**	$\lvert AF_n\rvert = \left\lvert\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right\rvert$	AF_n = normalized array factor N = number of elements	2(d)
Main beam	$\cos\theta_m = -\frac{\delta}{\beta d}$	θ_m = direction of the main beam	2(d)
Nulls	$\theta_k = \cos^{-1}\left[\frac{1}{\beta d}\left(-\delta \pm \frac{2k\pi}{N}\right)\right]$	θ_k = direction of null k k = 1, 2, …	2(d)
Broadside beamwidths	FNBW $\approx \frac{2\lambda}{Nd}$, HPBW $\approx 0.886\frac{\lambda}{Nd}$	λ = wavelength (m)	2(d)

Table 6: Arrays

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7. Links, radar and remote sensing:

Title	Equation	Notation	Ref.
★ **Friis formula**	$\frac{P_r}{P_t} = \frac{A_{et}A_{er}}{r^2\lambda^2} = G_tG_r\left(\frac{\lambda}{4\pi r}\right)^2$	P_r, P_t = received and transmitted power (W) A_{et}, A_{er} = effective apertures (m^2) r = distance (m) λ = wavelength (m) G_t, G_r = gains of the transmitting and receiving antennas	2(c), 6(c)
★ **Noise power**	$N = kT_{sys}B$	N = noise power (W) k = Boltzmann's constant (J/K) T_{sys} = system temperature (K) B = bandwidth (Hz)	2(c)
★ **Link SNR**	$\frac{S}{N} = \frac{P_tG_tG_r}{kT_{sys}B}\left(\frac{\lambda}{4\pi r}\right)^2$	S = received signal power (W)	2(c)
★ **Radar range**	$R_{max} = \left[\frac{P_tG^2\lambda^2\sigma}{(4\pi)^3P_{min}}\right]^{1/4}$	R_{max} = largest range (m) G = antenna gain σ = radar cross-section (m^2) P_{min} = smallest detectable power (W)	2(b)
Brightness temperature	$T_B = e\,T_{phys}$	T_B = brightness temperature (K) e = emissivity T_{phys} = physical temperature (K)	2(a)
Received noise power	$P = kT_AB$	P = noise power received (W) T_A = antenna temperature (K)	2(a)

Table 7: Links, radar and remote sensing

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Reflection coefficient	$\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$
Standing wave ratio	$S = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$
Input impedance	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$
Short and open input	$Z_{sc} = jZ_0 \tan\beta l,\ Z_{oc} = -jZ_0 \cot\beta l$
Quarter-wave transformer	$Z_1 = \sqrt{Z_0 R_L}$
High-pass ABCD	$A = 1 - \dfrac{1}{\omega^2 LC},\ B = \dfrac{1}{j\omega C},\ C = \dfrac{1}{j\omega L},\ D = 1$
Barkhausen condition	$\lvert A\beta\rvert = 1,\ \angle A\beta = 0°$
Directivity from aperture	$D = \dfrac{4\pi A_e}{\lambda^2}$
Directivity	$D = \dfrac{4\pi U_m}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$
Paraboloid taper	$\dfrac{S(\theta)}{S(0)} = \left(\dfrac{1+\cos\theta}{2}\right)^2,\ \dfrac{E(\theta)}{E(0)} = \dfrac{1+\cos\theta}{2}$
Horn directivity	$D \approx \dfrac{7.5 A_p}{\lambda^2}$
Array factor	$\lvert AF_n\rvert = \left\lvert \dfrac{\sin(N\psi/2)}{N\sin(\psi/2)} \right\rvert$
Friis formula	$\dfrac{P_r}{P_t} = \dfrac{A_{et}A_{er}}{r^2\lambda^2} = G_t G_r \left(\dfrac{\lambda}{4\pi r}\right)^2$
Noise power	$N = kT_{sys}B$
Link SNR	$\dfrac{S}{N} = \dfrac{P_t G_t G_r}{kT_{sys}B}\left(\dfrac{\lambda}{4\pi r}\right)^2$
Radar range	$R_{max} = \left[\dfrac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
Γ_L	reflection coefficient at the load	Reflection coefficient
Z_L	load impedance (Ω)	Reflection coefficient
Z_0	characteristic impedance of the line (Ω)	Reflection coefficient
S	standing wave ratio, V_{max}/V_{min}	Standing wave ratio
Γ	reflection coefficient	Standing wave ratio
d_{max}	distance of the first voltage maximum from the load (m)	Voltage maximum
θ	angle of Γ_L (rad)	Voltage maximum
λ	wavelength on the line (m)	Voltage maximum
Z_{in}	input impedance (Ω)	Input impedance
j	$\sqrt{-1}$	Input impedance
β	phase constant (rad/m)	Input impedance
l	line length (m)	Input impedance
V, I	voltage and current on the line (V, A)	Short-circuited line
V_0^+	incident amplitude (V)	Short-circuited line
z	position, the load at $z = 0$ (m)	Short-circuited line
Z_{sc}, Z_{oc}	input impedances with the end shorted and open (Ω)	Short and open input
Z_1	section impedance (Ω)	Quarter-wave transformer
R_L	real load (Ω)	Quarter-wave transformer
t	$\tan \beta d$	Stub position
d	distance from the load (m)	Stub position
X_L	load reactance (Ω)	Stub position
l	stub length (m)	Short-circuited stub
Y_0	$1/Z_0$, line admittance (S)	Short-circuited stub
B	susceptance to cancel (S)	Short-circuited stub
Z	impedance of the resistor (Ω)	High-frequency resistor
ω	angular frequency (rad/s)	High-frequency resistor
L	each lead's inductance (H)	High-frequency resistor
R	resistance (Ω)	High-frequency resistor
C_a	end capacitance (F)	High-frequency resistor
f_c	corner where C_a starts to short R (Hz)	Resistor corners
f_0	series resonance of the leads with C_a (Hz)	Resistor corners
V_1, I_1	voltage and current at port 1	ABCD definition
V_2, I_2	voltage and current at port 2	ABCD definition
A, B, C, D	ABCD parameters	ABCD definition
C	series capacitance (F)	High-pass ABCD
L	shunt inductance (H)	High-pass ABCD

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Symbol	Meaning	First used in
Γ_{in}	reflection at the input	Input reflection
S_{11}, S_{12}, S_{21}, S_{22}	S-parameters of the transistor	Input reflection
C_L, R_L	centre and radius of the output stability circle	Output stability circle
Δ	$S_{11}S_{22} - S_{12}S_{21}$	Output stability circle
S_{11}^*, S_{22}^*	complex conjugates of S_{11}, S_{22}	Output stability circle
C_S, R_S	centre and radius of the input stability circle	Input stability circle
R_{in}, X_{in}	input resistance and reactance of the device (Ω)	Negative-resistance oscillator
R_L, X_L	load resistance and reactance (Ω)	Negative-resistance oscillator
A	gain	Barkhausen condition
β	feedback fraction	Barkhausen condition
D	directivity	Directivity from aperture
A_e	effective aperture (m^2)	Directivity from aperture
P_{rad}	radiated power (W)	Power and beam area
U	radiation intensity (W/sr)	Power and beam area
Ω	solid angle (sr)	Power and beam area
Ω_A	beam area (sr)	Power and beam area
U_m	largest radiation intensity (W/sr)	Power and beam area
r_1, r_2	inner and outer limits of the radiating near field (m)	Field regions
D	largest dimension of the antenna (m)	Field regions
E_ϕ	electric field (V/m)	Magnetic dipole E
η	wave impedance of free space (377 Ω)	Magnetic dipole E
k	$2\pi/\lambda$ (rad/m)	Magnetic dipole E
I	loop current (A)	Magnetic dipole E
S	loop area (m^2)	Magnetic dipole E
r	distance (m)	Magnetic dipole E
θ	angle from the loop axis	Magnetic dipole E
H_r	radial magnetic field (A/m)	Magnetic dipole H_r
H_θ	θ component of the magnetic field (A/m)	Magnetic dipole H_θ
f	focal length (m)	Paraboloid shape
$S(\theta)$	power density at the aperture (W/m^2)	Paraboloid taper
$E(\theta)$	field at the aperture (V/m)	Paraboloid taper
A_p	mouth area (m^2)	Horn directivity
F	polarization match factor	Polarization match
γ	half the angle between the two polarizations on the Poincaré sphere	Polarization match
ψ	phase difference between neighbouring elements (rad)	Phase difference
d	spacing (m)	Phase difference
θ	angle from the array axis	Phase difference

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Symbol	Meaning	First used in
δ	feed phase (rad)	Phase difference
AF_n	normalized array factor	Array factor
N	number of elements	Array factor
θ_m	direction of the main beam	Main beam
θ_k	direction of null k	Nulls
P_r, P_t	received and transmitted power (W)	Friis formula
A_{et}, A_{er}	effective apertures (m^2)	Friis formula
G_t, G_r	gains of the transmitting and receiving antennas	Friis formula
N	noise power (W)	Noise power
T_{sys}	system temperature (K)	Noise power
B	bandwidth (Hz)	Noise power
S	received signal power (W)	Link SNR
R_{max}	largest range (m)	Radar range
G	antenna gain	Radar range
σ	radar cross-section (m^2)	Radar range
P_{min}	smallest detectable power (W)	Radar range
T_B	brightness temperature (K)	Brightness temperature
e	emissivity	Brightness temperature
T_{phys}	physical temperature (K)	Brightness temperature
P	noise power received (W)	Received noise power
T_A	antenna temperature (K)	Received noise power

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